

★ WEEK 16 · PART 1 of 3

Three Families, and How to Choose

Classical, State-Space, and Bayesian - the Whole Course, Sorted

Three questions · three kinds of forecast · one decision-driven choice

The final week opens by stepping back. Everything you learned sorts into three model families, each answering a different question. There is no universal best model - the right one depends on the decision you face.

1

PART 1 · THE THREE FAMILIES

One Course, Three Questions

The forecasting competitions that taught the field to test, the three model families and what each one answers, the three kinds of forecast they produce, and how to let the decision choose the model.

The Forecasting Competitions — Part 1

How the field learned to stop arguing and start testing

For years, forecasters argued about which method was best in theory. Then the M-competitions put dozens of methods on the same series and judged them on data they had never seen. The surprise: elaborate models often did NOT beat simple ones.

What the competitions did

- Many methods forecast the very same series.
- Each was graded on held-out future data.
- The scoreboard was public and out-of-sample.

The uncomfortable finding

- Sophistication did not guarantee accuracy.
- Simple methods were often hard to beat.
- Theory alone couldn't settle the argument.

The Forecasting Competitions — Part 2

The lesson: let unseen data decide

The lasting lesson is the backbone of this week. Don't crown a model on elegance or in-sample fit. Fit on the past, then grade every model on the future it never saw - and let that honest test decide.

The lesson

- Out-of-sample accuracy is the real judge.
- In-sample fit can flatter a complex model.
- Compare fairly, on the same held-out data.

How this week uses it

- Apply all three families to one asset.
- Grade them on data none of them saw.
- Let the scoreboard, not taste, decide.

The Whole Course Was One Arc

Three families, three questions

1

Classical asked: what is the mean and variance?

ARMA and GARCH (Weeks 2-6) model the average return and how its volatility clusters - a point forecast and a risk level.

2

State-space asked: what is hidden and changing?

Kalman filters and HMMs (Weeks 8-12) track a moving hidden state or regime behind the noisy data.

3

Bayesian asked: how uncertain is all of it?

Priors and MCMC (Weeks 13-15) turn point estimates into full distributions - honest doubt over states and parameters.

Part 1 in one line: three families answer three different questions - and no single one wins everywhere.

The Three Families at a Glance

Classical models the mean and variance; state-space tracks the hidden regime; Bayesian delivers the full distribution. The pairs thread runs through all three, and out-of-sample testing compares them on the same asset

The Three Families at a Glance

Each family answers a different question about one asset - and the whole course is the arc across them.



Tied together two ways

The pairs-trading thread runs through all three (find the pair, track the hedge ratio, trade with uncertainty), and out-of-sample testing is how we compare them fairly on the very same asset.

Three Families, Three Questions

Why they are complements, not rivals

The families feel like competitors, but they answer different questions. One wants the best single number; one wants to know which hidden regime we are in; one wants the whole range of what could happen. A real decision usually needs one of these more than the others

Different questions

- Classical: the mean and variance of returns.
- State-space: the hidden state or regime.
- Bayesian: the full uncertainty around it all.

So different answers

- A point forecast and a volatility.
- A tracked state or a regime probability.
- A predictive distribution with credible bands.

Classical: ARMA and GARCH

Modeling the mean and the variance of returns

The classical family models returns directly: ARMA captures the predictable part of the mean, and GARCH captures how volatility clusters and persists. Its output is a point forecast of the return and a forecast of its variance.

What it captures

- ARMA: the autocorrelation in the mean.
- GARCH: volatility that clusters over time.
- A parsimonious description of the series.

What it gives

- A point forecast for the next return.
- A forecast of the variance around it.
- Simple, fast, and often hard to beat.

Classical, Across the Course

Where you already used it

The tools you built

- Stationarity and the ACF/PACF (Weeks 1-2).
- AR, MA, ARMA, and ARIMA models (Weeks 3-4).
- GARCH for clustered volatility (Weeks 5-6).

Its strengths and limits

- Strength: clean, interpretable, efficient.
- Limit: assumes fixed parameters.
- Limit: no hidden state, no honest band.

State-Space: Kalman and HMM

Tracking what is hidden and changing

The state-space family assumes something unobserved drives the data - a slowly drifting level, a time-varying beta, a hidden regime. The Kalman filter tracks a continuous hidden state; the HMM tracks a discrete regime. The output is that tracked state.

What it captures

- A hidden state that evolves over time.
- Continuous drift (Kalman) or regimes (HMM).
- Parameters that move, not stay frozen.

What it gives

- A filtered estimate of the hidden state.
- A regime label or probability at each date.
- Adaptation the classical family lacks.

State-Space, Across the Course

Where you already used it

The tools you built

- State-space form and the Kalman filter (Weeks 8-9).
- Hidden Markov models for regimes (Week 10).
- SDEs and short-rate models (Weeks 11-12).

Its strengths and limits

- Strength: tracks moving states and regimes.
- Limit: usually a point estimate of the state.
- Limit: parameters still taken as known.

Bayesian: Full Uncertainty

Parameter doubt and predictive distributions

The Bayesian family stops pretending the parameters are known. It places priors, infers a full posterior, and produces a predictive distribution. Its output is not a line but a fan - the honest range of what could happen, states and parameters included.

What it captures

- Uncertainty about the parameters themselves.
- A joint posterior over states and parameters.
- The two layers of doubt from Week 15.

What it gives

- A predictive distribution, not one number.
- Credible bands on every quantity.
- Risk you can size a decision against.

Bayesian, Across the Course

Where you already used it

The tools you built

- Bayesian inference and conjugacy (Week 13).
- MCMC and HMC-NUTS sampling (Week 14).
- Bayesian state-space and switching (Week 15).

Its strengths and limits

- Strength: honest, decision-ready uncertainty.
- Limit: heavier to compute and to check.
- Limit: needs care with priors and diagnostics.

The Three Families in One Table

The synthesis at a glance

Family	Question it answers	Tools	Forecast
Classical	Mean and variance of returns	ARMA / GARCH	A point + a variance
State-Space	Which hidden state or regime	Kalman / HMM	A tracked state
Bayesian	How uncertain is it all	Priors / MCMC	A full distribution

Read each row as a decision: pick the family whose forecast type matches what your decision needs.

What Each Family Actually Forecasts

Point, regime, or full distribution

The cleanest way to tell the families apart is by the shape of their forecast. Classical gives one number per day. State-space gives the regime you are probably in. Bayesian gives the whole distribution of what could happen next.

Three shapes of answer

- A single line: the point forecast.
- A shaded regime: which state lies ahead.
- A fan: the full predictive distribution.

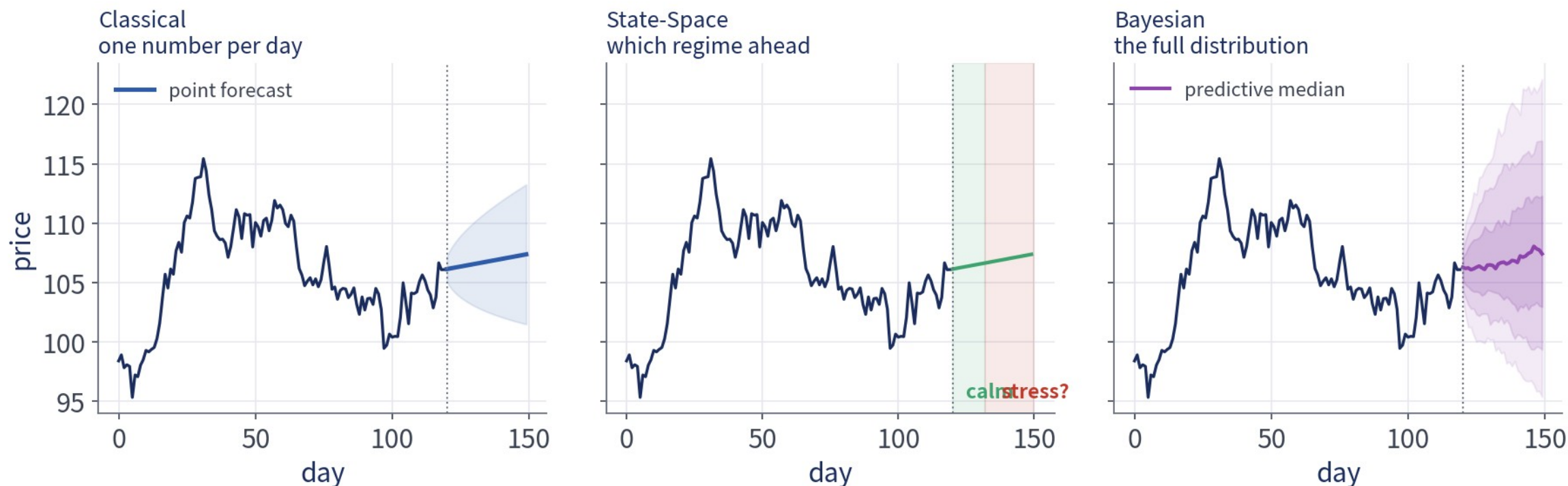
Why the shape matters

- It determines what decisions you can make.
- A number can't tell you the tail risk.
- A distribution can't be beaten for sizing risk.

The Same Asset, Forecast Three Ways

One asset's future as a point forecast (Classical), as a regime call (State-Space), and as a full predictive fan (Bayesian). None is 'the truth' - each answers a different question, and each fits a different decision.

The same asset, forecast three ways - match the forecast to your decision



Match the Forecast to the Decision

The forecast shape you need depends on the choice

If your decision is...

- 'What return do I book?' - you need a point.
- 'Is it safe to be on?' - you need the regime.
- 'How much can I lose?' - you need the distribution.

...then reach for

- Classical, for the single best estimate.
- State-space, for the regime probability.
- Bayesian, for the tail and the credible band.

There Is No Universal Best Model

The right choice depends on the decision

This is the central idea of the week. Asking 'which family is best?' is the wrong question - the competitions showed no model wins everywhere. The right question is: what does THIS decision need? Then the choice follows.

Why no model wins always

- Each is best at its own question.
- Complexity does not guarantee accuracy.
- The data and the decision differ every time.

So how you decide

- Start from the decision, not the model.
- Match the forecast shape to the need.
- When in doubt, run all three and test.

How to Choose in Practice

A short checklist

Ask three questions

- What exactly must the forecast support?
- What shape of answer does that need?
- How much does being wrong cost?

Then act

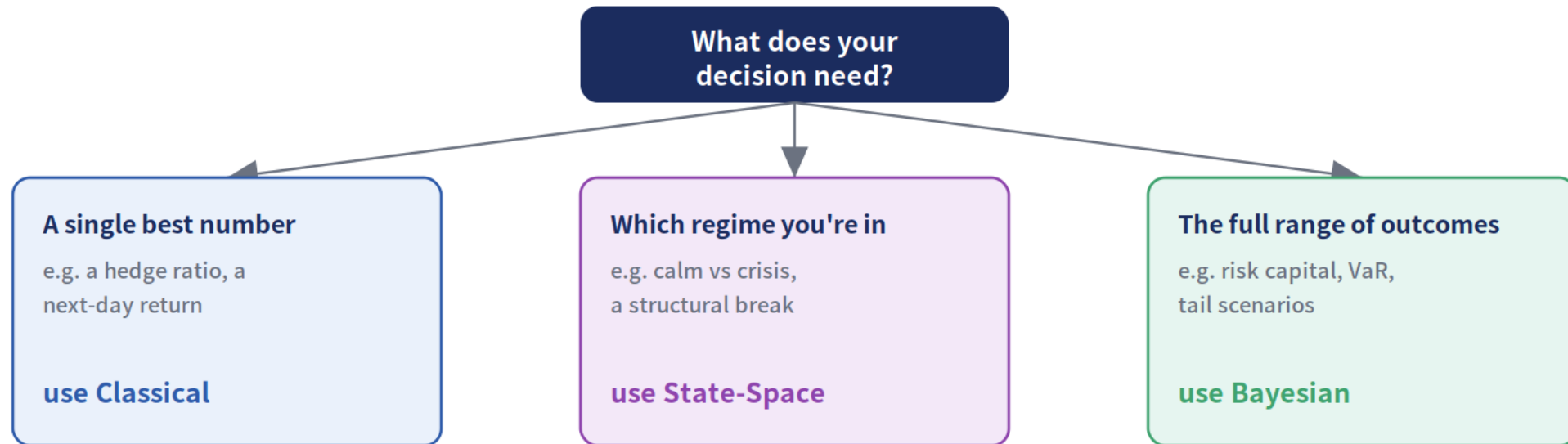
- Pick the family whose output matches.
- If the stakes are high, run all three.
- Let out-of-sample accuracy break ties.

Let the Decision Choose the Model

Ask what the decision needs: a single best number points to Classical; knowing the regime points to State-Space; the full range of outcomes points to Bayesian. And when it matters, run all three and let the test decide

Let the Decision Choose the Model

There is no universal best model. Ask what your decision actually needs, then pick the family.



And often: run all three, then let out-of-sample accuracy break the tie.

The families are complements, not rivals - the capstone compares them fairly on one asset.

The Time-Series Toolbox, in Trading

This is what all the models were for. ARMA sets the entry direction, GARCH the size, cointegration the pair itself, Kalman the hedge ratio, HMM a regime gate, OU the holding period, and Bayesian the conviction. Part 3 assembles them into one trade.

The Time-Series Toolbox, in Trading

Every model you learned maps to a concrete trading job - a signal, a size, a gate, or a holding period.

ARMA (W3-4)

forecasts the mean

-> entry direction: which way the spread moves next

GARCH (W5-6)

forecasts volatility

-> position size: scale inversely with risk; set VaR

Cointegration (W7)

the long-run spread

-> the pair itself: define the spread and its bands

Kalman (W9)

time-varying hedge ratio

-> the long/short ratio, adjusted in real time

HMM (W10)

the regime probability

-> a gate: cut or exit when a crisis regime is likely

OU process (W11)

mean-reversion speed

-> holding period and entry threshold (the half-life)

Bayesian SS (W15)

the credible band

-> conviction sizing: bet less when the band is wide

What Parts 2 and 3 Add

From choosing to testing to delivering

Part 2 - the fair comparison

- Apply all three to one asset.
- Hold-out testing and walk-forward.
- Metrics a trading desk actually uses.

Part 3 - the final project

- The capstone, step by step.
- From comparison to a trading decision.
- Presenting it, and where you go next.

Three Families on One Asset

Direct Claude Code to set up the comparison

GOAL Fit one representative from each family to a single asset and compare their forecast shapes.

[INPUT] Prompt · Claude Code

Write and run `three_families.py` that:

1. loads one asset's returns and splits into a training period and a held-out test period;
2. fits a Classical model (ARMA + GARCH), a State-Space model (Kalman or HMM), and a Bayesian model (a simple posterior/predictive);
3. produces each family's forecast for the test period: a point, a regime, and a distribution;
4. plots all three against the realized path.

Note which decision each output would best serve.

How to work with it

- Same asset, same split.
- Three shapes of forecast.
- None is 'the truth'.
- Part 2 will grade them fairly.

Deliverable: `three_families.py` + a point, a regime, and a distribution forecast for one asset.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1 What question does each of the three families answer?

**Mean/variance,
hidden state, full
uncertainty**

Q2 What forecast shape does each produce?

**A point, a regime, a
distribution**

Q3 How should you choose among them?

**By what the decision
needs, then test**

All three come from the three questions, the three forecast shapes, and the decision-first rule.

Three Misconceptions to Avoid

Where synthesis intuition slips

Common myths

- "The most sophisticated model is the best."
- "A great in-sample fit means great forecasts."
- "One family should be used for everything."

The reality

- The competitions showed simple can win.
- In-sample fit can flatter and mislead.
- Match the family to the decision at hand.

Three Things to Carry Forward

Part 1 in one screen

1

The course sorts into three families

Classical models the mean and variance; state-space tracks the hidden state or regime; Bayesian delivers the full distribution. Each answers a different question about the same asset.

2

The forecast shape is the key difference

A point, a regime, or a full distribution. What you can decide depends on which shape you hold, so match the forecast to the decision you face.

3

There is no universal best model

The competitions taught the field to test, not to argue. Start from the decision, pick the family whose output fits, and when it matters, run all three and let unseen data decide.

Next — Part 2: apply all three to one asset and compare them fairly, out of sample.

★ WEEK 16 · PART 1 · PRACTICE

Choosing a Model, Worked by Hand

Grade Forecasts Out-of-Sample; Match the Shape to the Decision

Naive vs AR(1) on a scoreboard - and a point vs a distribution

Part 1's two big ideas, with the arithmetic shown. We grade two forecasts on held-out data and let the numbers pick a winner, then compute a tail risk a point forecast can never reveal.

1P

PART 1 · APPLICATIONS

Two Cases, Computed by Hand

Case A: forecast a return two ways, grade both on four held-out days, and let RMSE decide. Case B: turn a mean and a volatility into a tail probability and a worst-case loss. Then a Claude Code lab.

What Part 1 Told Us

Now we make it concrete - with numbers

1

Three families, three forecast shapes

A point, a regime, or a full distribution. Which one you need depends on the decision in front of you.

2

There is no universal best model

The competitions taught the field to test, not argue. We will grade two forecasts on held-out data and let the numbers decide.

3

Match the shape to the decision

A point forecast answers 'what return?'; only a distribution answers 'how much can I lose?'. We compute both by hand.

Everything here is a calculator exercise: four held-out days, one mean, one volatility.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the key numbers by hand, step by step.
- Then read off the decision it supports.

What you should watch

- How a held-out test crowns a winner.
- How a point forecast stays silent on risk.
- How the distribution answers the risk question.

Grade the Forecasts Out-of-Sample

Let the held-out data crown the winner

A desk needs a next-day return forecast for a signal. Two candidates: the naive forecast (tomorrow equals today) and an AR(1) model. Which is better? We do not argue - we grade both on four days neither saw, and let the errors decide.

The two candidates

- Naive: repeat the last observed return.
- AR(1): scale the last return by a coefficient.
- The naive forecast is the bar to beat.

What we'll do by hand

- Forecast four held-out days each way.
- Compute the errors and their squares.
- Compare RMSE and MAE, and pick a winner.

Naive vs AR(1)

The rules we will grade

$$\text{naive: } \hat{r}_t = r_{t-1} \qquad \text{AR(1): } \hat{r}_t = \phi r_{t-1}, \quad \phi = 0.5$$

The naive forecast

- Tomorrow's return equals today's.
- No parameters, no fitting.
- Surprisingly hard to beat.

The AR(1) forecast

- Scale the last return by $\phi = 0.5$.
- ϕ was estimated on the training data.
- More sophisticated - but is it better?

The Held-Out Test: Four Days

Forecast each day two ways

Day	$r(t-1)$	actual $r(t)$	naive fc	AR(1) fc
1	+1.0	+0.4	1.0	0.5
2	+0.4	-0.6	0.4	0.2
3	-0.6	+0.8	-0.6	-0.3
4	+0.8	-0.2	0.8	0.4

Goal: score each forecast by how far it lands from the actual. $AR(1) \text{ fc} = 0.5 \times r(t-1)$; $naive \text{ fc} = r(t-1)$.

The Errors

Actual minus forecast, then squared

naive errors: $-0.6, -1.0, +1.4, -1.0 \Rightarrow \text{squared sum} = 4.32$
AR(1) errors: $-0.1, -0.8, +1.1, -0.6 \Rightarrow \text{squared sum} = 2.22$

The AR(1) errors are smaller day by day, so its squared-error sum (2.22) is far below the naive's (4.32).

RMSE, MAE, and the Winner

Turn errors into a score

$$\text{RMSE}_{\text{naive}} = \sqrt{4.32/4} = 1.04 \quad \text{RMSE}_{\text{AR}} = \sqrt{2.22/4} = \mathbf{0.74}$$

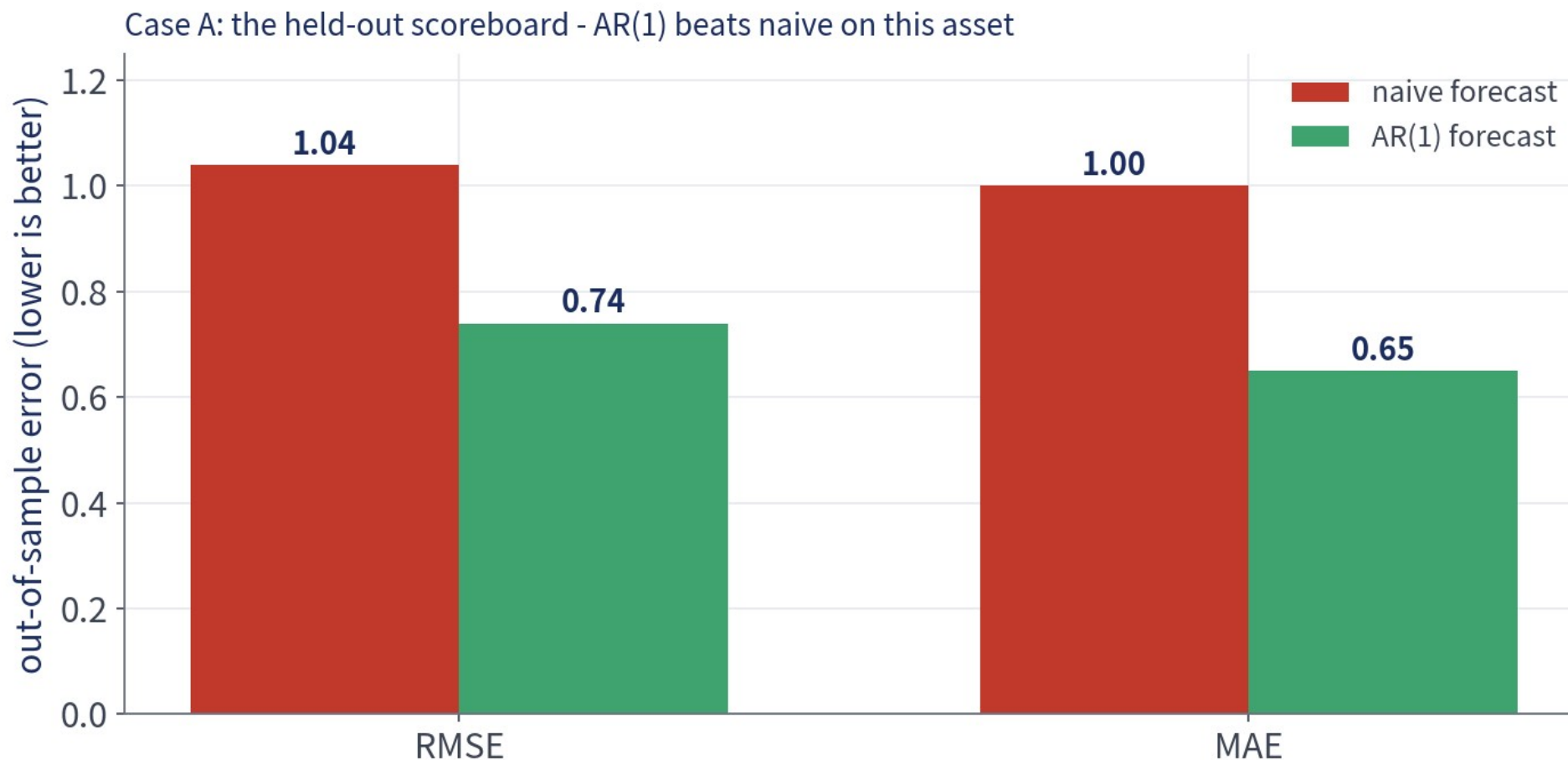
$$\text{MAE}_{\text{naive}} = 4.0/4 = 1.00 \quad \text{MAE}_{\text{AR}} = 2.6/4 = \mathbf{0.65}$$

$\Rightarrow$ AR(1) wins the held-out test - on this asset

On this asset AR(1) wins - but the point is the METHOD: on another asset the naive might win, which is exactly why you test.

The Held-Out Scoreboard

The same numbers as a picture: on both RMSE and MAE, the AR(1) forecast sits below the naive one for this asset. Lower is better - and this held-out comparison, not in-sample fit, is what settles the choice.



Why the Naive Forecast Is the Benchmark

Beat it, or don't bother

The naive forecast

- Costs nothing and needs no fitting.
- For many financial series it is very strong.
- It is the honest bar every model must clear.

The discipline

- Always report your model's error next to naive.
- If you can't beat naive, prefer the simpler one.
- Sophistication is only worth it if it scores better.

What the Test Decides

Don't trust a fit you can't defend

The wrong way

- Crown the model with the best in-sample fit.
- Assume the fancier model must forecast better.
- Never check it on data it hasn't seen.

The right way

- Fit on the past, grade on the held-out future.
- Compare against the naive benchmark.
- Let the out-of-sample score pick the winner.

When a Point Isn't Enough

A risk decision needs the whole distribution

Now a different decision: how much could this position lose next week? A point forecast of +0.5% sounds reassuring - but it says nothing about the downside. For a risk decision you need the distribution, and we compute its tail by hand.

The decision

- Set risk capital for a bad week.
- You need the tail, not the average.
- A single number cannot answer this.

What we'll do by hand

- Take the mean and the volatility.
- Compute the 5% worst-case return.
- Compute the probability of a big loss.

A Point vs a Distribution

Same forecast, two very different answers

The point forecast says

- 'Next week: about +0.5%.'
- Comforting, and completely silent on risk.
- It hides the whole left tail.

The distribution says

- '+0.5% on average, but with this spread.'
- It names the worst case and its odds.
- That is what a risk limit is built on.

The Forecast: Mean and Vol

Two numbers to build a distribution

Forecast input	Value
mean return (μ)	+0.5%
volatility (σ)	2.0%

Assume next-week returns are roughly normal with this mean and volatility. Goal: find the 5% worst case and the odds of a big loss.

The 5% Worst Case

One line from mean and vol

$$\begin{aligned} \text{forecast: mean } \mu &= +0.5\%, \text{ vol } \sigma = 2.0\% \\ 5\% \text{ worst case} &= \mu - 1.645 \sigma = 0.5 - 1.645(2.0) = -\mathbf{2.79\%} \end{aligned}$$

There is a 5% chance next week's return is worse than -2.79%. The point forecast of +0.5% never hinted at this.

The Odds of a Big Loss

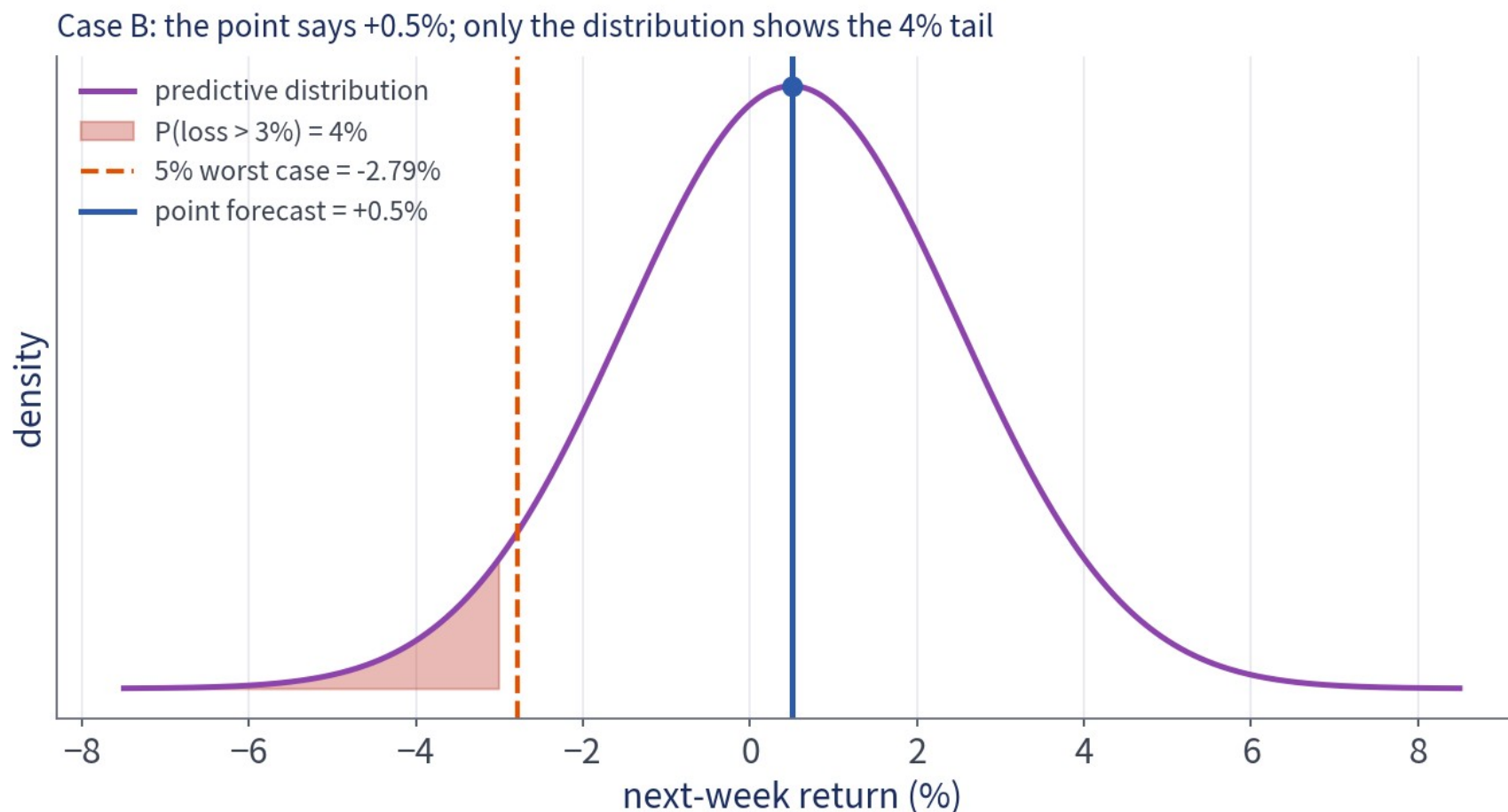
A z-score and a table lookup

$$\begin{aligned} P(r < -3\%) &= P\left(Z < \frac{-3-0.5}{2.0}\right) = P(Z < -1.75) \\ &= \Phi(-1.75) = 0.040 = 4\% \end{aligned}$$

Standardize the threshold to a z-score, then read the normal table: about a 4% chance of losing more than 3% in the week.

The Point Hides the Tail

The point forecast is one dot at +0.5%. The predictive distribution shows the whole picture: a 5% worst case near -2.79% and a shaded 4% chance of losing more than 3%. Only the distribution answers the risk question



Match the Shape to the Decision

Which family, for which question

If the decision is...

- 'What return do I book?' - a point is fine.
- 'Am I in a safe regime?' - you need the regime.
- 'How much can I lose?' - you need the distribution.

...then the family is

- Classical: the point forecast from Case A.
- State-space: the regime probability.
- Bayesian: the predictive tail from Case B.

What the Distribution Buys You

Risk you can size against

With only the point

- You see +0.5% and feel safe.
- You cannot set a risk limit from it.
- The tail catches you by surprise.

With the distribution

- You quote the 5% worst case: -2.79%.
- You size risk capital to the tail.
- You are ready for the bad week.

The Common Pattern

Two calculations, one discipline

Both cases, computed by hand, enforce the same discipline. Don't assume a model is best - grade it on held-out data. Don't accept a point when the decision needs the tail - compute the distribution. The right answer always starts from the decision.

The trap

- Trusting in-sample fit or a lone number.
- Assuming complex beats simple.
- Answering a risk question with an average.

The fix

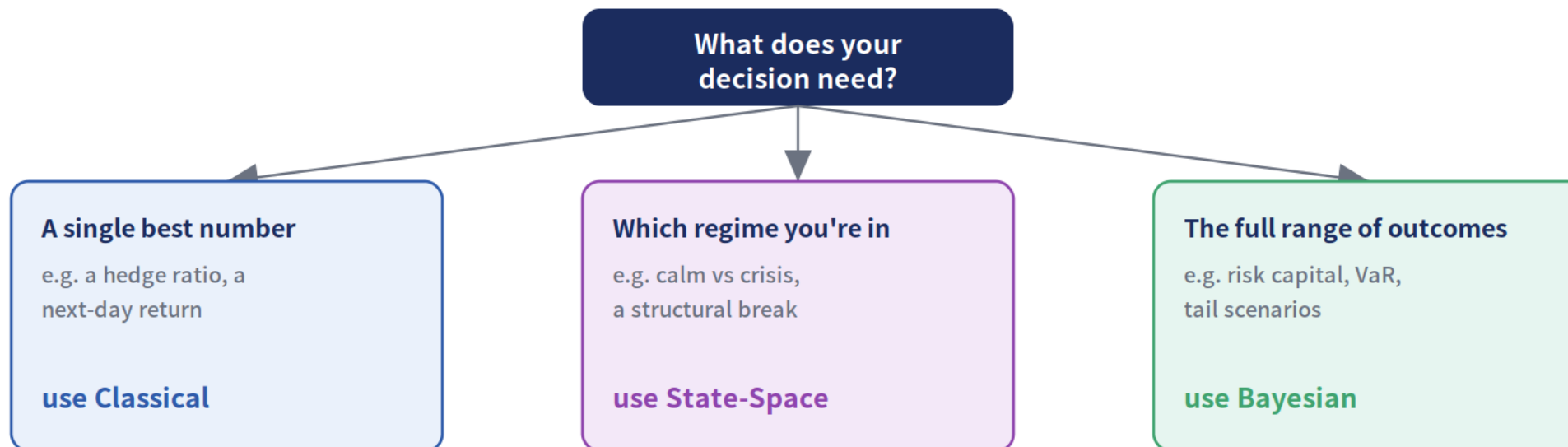
- Grade every model out-of-sample.
- Beat the naive benchmark or simplify.
- Match the forecast shape to the decision.

Let the Decision Choose the Model

Case A gave a point for a signal; Case B gave a distribution for risk. The map is the same rule: a single number points to Classical, the regime to State-Space, the full range to Bayesian - and out-of-sample testing breaks ties

Let the Decision Choose the Model

There is no universal best model. Ask what your decision actually needs, then pick the family.



And often: run all three, then let out-of-sample accuracy break the tie.

The families are complements, not rivals - the capstone compares them fairly on one asset.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both worked examples in code, then extend them to a real hold-out.

[INPUT] Prompt · Claude Code

Write and run `part1_practice.py` that:

CASE A - grade forecasts:

- * take the 4-day table; form naive and AR(1) forecasts; print each error and squared error;
- * compute RMSE and MAE for both; name the winner;
- * then repeat on a real asset with a train/test split and compare to the naive benchmark.

CASE B - point vs distribution:

- * from $\mu = 0.5\%$, $\sigma = 2\%$, print the 5% worst-case return and $P(\text{loss} > 3\%)$;
- * plot the point against the full distribution.

Match every printed number to the slides.

Check against the slides

- RMSE naive 1.04, AR 0.74.
- MAE naive 1.00, AR 0.65.
- 5% worst case = -2.79%.
- $P(\text{loss} > 3\%) = 4\%$.

Deliverable: `part1_practice.py` whose printed numbers match the hand calculations, then the charts.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 How do you decide which forecast is better?

**Lower out-of-sample
RMSE/MAE**

Q2 Why is the naive forecast important?

**It's the benchmark
every model must
beat**

Q3 Why can't a point forecast set risk capital?

**It says nothing about
the tail**

All three come from the held-out scoreboard, the naive bar, and the tail calculation.

Three Pitfalls to Avoid

Where model choice slips

Common mistakes

- "The best in-sample fit is the best model."
- "A fancier model must forecast better."
- "A point forecast is enough for any decision."

The reality

- In-sample fit flatters; grade out-of-sample.
- Beat the naive benchmark, or simplify.
- Risk decisions need the whole distribution.

Two Calculations to Remember

Part 1 applications in one screen

1

Grade forecasts out-of-sample

Naive and AR(1) scored RMSE 1.04 vs 0.74 on four held-out days. AR(1) won here - but the discipline is what matters: fit on the past, grade on the future, and beat the naive benchmark or simplify.

2

A risk decision needs the distribution

A point forecast of +0.5% is silent on risk. The distribution gives a 5% worst case of -2.79% and a 4% chance of losing more than 3% - the numbers a risk limit is actually built on.

3

Start from the decision

Match the forecast shape - point, regime, or distribution - to what the decision needs, then let out-of-sample testing settle any remaining choice.

Next — Part 2: the fair comparison, applying all three families to one asset and testing them.

★ WEEK 16 · PART 2 of 3

The Fair Comparison: Out-of-Sample Testing

One Asset, Three Families, One Honest Grade

Hold-out testing · walk-forward · the metrics a desk actually uses

Choosing a family is only half the work; proving it is the rest. This part builds the capstone comparison: fit all three families on the past, grade them on the future they never saw, and score them the way a trading desk would.

2

PART 2 · THE FAIR COMPARISON

Fit on the Past, Grade on the Future

The capstone pipeline, hold-out and walk-forward testing, the look-ahead bias that fakes great results, the metrics that matter in finance, and how to read the scoreboard without fooling yourself.

From Choosing to Proving

Part 1 chose; Part 2 tests

1

We have three families

Classical, state-space, and Bayesian - each with a forecast shape suited to a different decision.

2

But a choice needs evidence

The competitions taught us not to trust taste or in-sample fit. We must grade the families on data they never saw.

3

So we build a fair comparison

Apply all three to one asset, hold out a test period, and score them identically against a naive benchmark.

Part 2 in one line: a fair, out-of-sample comparison - the same asset, the same test, the same metrics.

Apply All Three to One Asset

The shape of the project, end to end

The capstone takes one asset and runs it through all three families. Each is fit on a training period and asked to forecast a held-out test period. Then a single, honest scoreboard compares them - and ties back to a trading decision.

What goes in

- One asset's return series.
- A clean split into train and test.
- One representative model per family.

What comes out

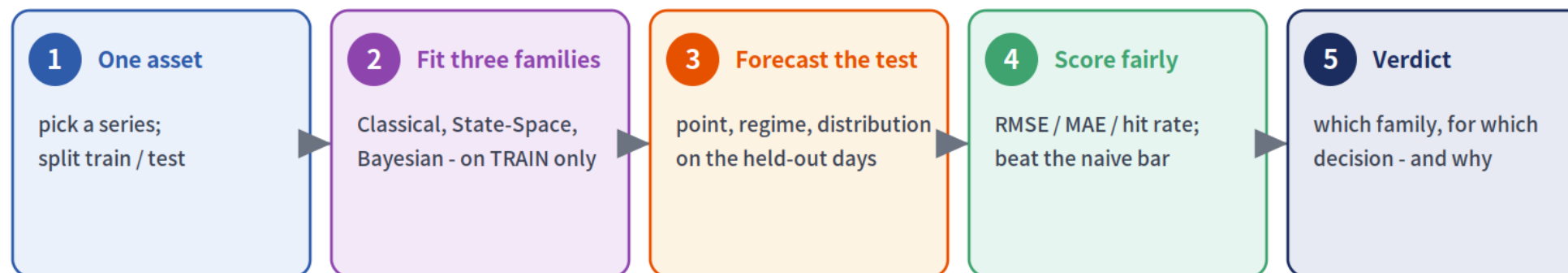
- A forecast from each family on the test.
- A scoreboard on identical metrics.
- A decision-driven verdict.

The Capstone as a Pipeline

One asset in; three families fit on the training data only; each forecasts the held-out test; all are scored on the same metrics against the naive bar; and the verdict follows the decision. Two rules keep it honest

The Capstone as a Pipeline

One asset in, three fitted families, one held-out grade, one honest verdict out.



The two rules that keep it honest

1. Fit on the past only - never let a model touch the test data before it is graded (no look-ahead bias).
2. Compare every family on the SAME held-out days, against the naive benchmark, with the same metrics.

The verdict is decision-driven: the winner is the family whose output your decision actually needs.

The Fair Comparison: Hold-Out Testing

Fit on the past, grade on the future it never saw

The heart of an honest comparison is the hold-out. Reserve the most recent slice of data as a test set, fit every model on the earlier part only, then grade each on the reserved slice. No model ever sees its own exam.

How it works

- Split the series into train and test.
- Fit each model on the training part.
- Forecast, then compare to the held-out truth.

Why it's fair

- Every model faces the same unseen test.
- In-sample fit can't flatter anyone here.
- The score reflects real forecasting skill.

Fit on the Past, Grade on the Future

The rule behind every honest test

Training

- Estimate all parameters on the past only.
- Let each model do its best on what it can see.
- This is where fitting happens - and stops.

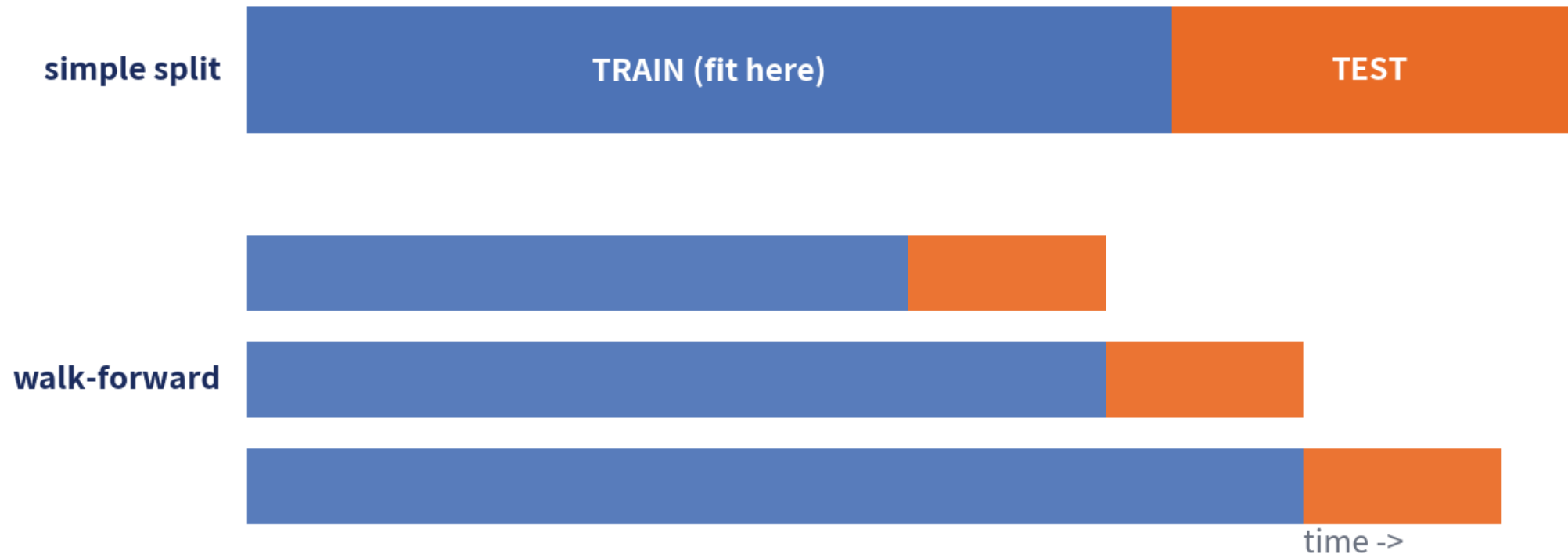
Testing

- Freeze the models; forecast the test period.
- Compare forecasts to the realized values.
- The grade is on data used for nothing else.

Train, Test, and Walk Forward

The simple version reserves one test slice at the end. Walk-forward does it repeatedly - fit, test one step, roll the window forward - so the grade averages over many honest out-of-sample forecasts rather than one lucky split

Fit on the past, grade on the future it never saw - once, or rolling forward



Walk-Forward Validation

One split can be lucky; many cannot

The idea

- Fit, forecast the next step, then roll forward.
- Re-fit as the window advances through time.
- Collect many out-of-sample forecasts.

Why it's better

- It averages over many test points, not one.
- It mimics how you'd actually trade in real time.
- A single split can flatter or punish by luck.

Look-Ahead Bias

The one mistake that fakes great results

Look-ahead bias is letting a model use information it could not have had at the time - a future value, a full-sample mean, a scaler fit on the test set. It produces gorgeous backtests that collapse in live trading. It is the single most common way to fool yourself.

How it sneaks in

- Scaling with statistics from the whole sample.
- Using a value that wasn't known yet.
- Tuning choices on the test set.

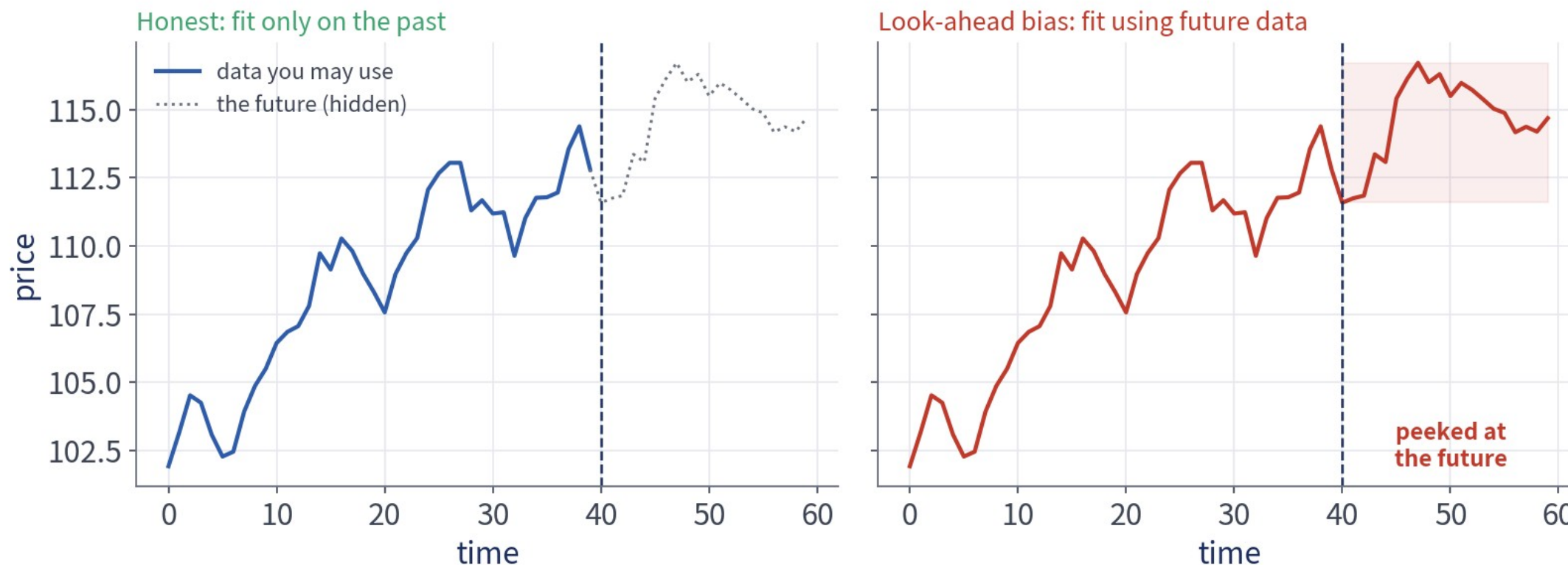
What it costs

- Backtests that look far too good.
- Results that vanish out of sample.
- Confidence in a strategy that doesn't work.

The Bias That Fakes Great Results

On the left, the honest setup uses only the past to fit. On the right, the model peeks at the future - and any statistic computed over the whole series does the same. Never let a model touch the data it will be graded on

The one bias that fakes great results: never let the model see its own test



How to Avoid Look-Ahead

A few habits that save you

Do this

- Compute every statistic on the training data only.
- Fit scalers and transforms on train, apply to test.
- Timestamp everything - use only what was known.

Never this

- Never normalize with full-sample mean or variance.
- Never tune hyperparameters on the test set.
- Never let a future bar leak into a past feature.

Metrics That Matter in Finance

Score forecasts the way a desk actually would

Not all errors are equal. A desk cares about the size of the error, whether the direction was right, and whether an honest uncertainty was reported. We use a small set of complementary metrics - and read them together, not one alone.

Three angles

- Size: how far off was the point forecast?
- Direction: did you get the sign right?
- Honesty: was the stated uncertainty calibrated?

Read them together

- A model can win on size but lose on direction.
- A tight point can hide an overconfident band.
- No single number tells the whole story.

Accuracy: RMSE and MAE

How far the point forecast lands

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_t (r_t - \hat{r}_t)^2}, \quad \text{MAE} = \frac{1}{n} \sum_t |r_t - \hat{r}_t|$$

What they measure

- RMSE: penalizes big misses heavily.
- MAE: the plain average miss.
- Both: lower is better, on the test set.

How to use them

- Report both next to the naive benchmark.
- RMSE if large errors hurt most.
- MAE if every miss counts the same.

Direction and P&L

Getting the sign right, and the money

$$\text{hit rate} = \frac{1}{n} \sum_t \mathbf{1}[\text{sign}(\hat{r}_t) = \text{sign}(r_t)], \quad \text{Sharpe of the signal's P\&L}$$

Why direction matters

- For a trading signal, the sign drives the P&L.
- A model can have low error yet wrong direction.
- The hit rate scores the sign directly.

The bottom line

- Ultimately, grade the signal's simulated P&L.
- A good forecast should make money out of sample.
- Report the strategy's Sharpe, not just the error.

Honesty: the Log Score

Reward a well-calibrated distribution

$$\text{log score} = \frac{1}{n} \sum_t \log p(r_t \mid \text{model}) \quad \text{rewards honest uncertainty, not just the point}$$

What it does

- It scores the whole predictive distribution.
- It rewards putting probability on what happens.
- It punishes overconfidence and vagueness alike.

Why it matters here

- RMSE ignores the band; the log score doesn't.
- It's how you grade the Bayesian family fairly.
- Calibrated uncertainty finally gets credit.

Which Metric for Which Decision

Score what the decision cares about

If you will...

- Book a number: use RMSE or MAE.
- Trade a direction: use hit rate and P&L.
- Size risk: use the log score / calibration.

The rule

- Grade on the quantity you'll actually act on.
- A model can win one metric and lose another.
- Choose the metric before you see the scores.

Reading the Forecast Scoreboard

Lower error wins - but mind the fine print

With the metrics in hand, the scoreboard looks simple: lowest error wins. But read the fine print. Are the gaps large enough to matter? Did every model beat naive? Is the winner winning on the metric your decision cares about?

What to check

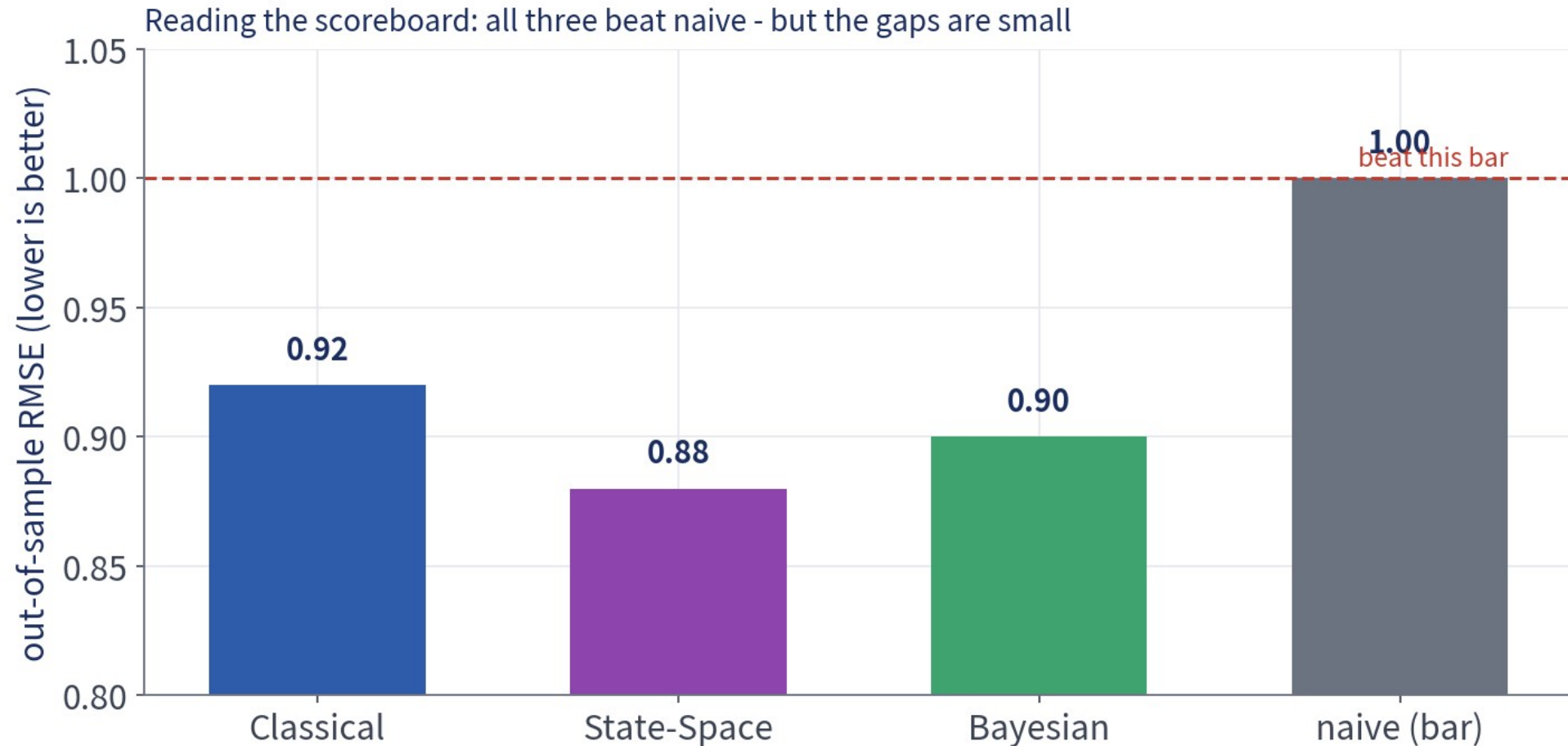
- Did each family clear the naive bar?
- Are the differences meaningful or noise?
- Which metric is the verdict resting on?

What to resist

- Don't crown a winner on a razor-thin gap.
- Don't ignore a family that fits your decision.
- Don't let one lucky split decide it.

Reading the Scoreboard

All three families beat the naive benchmark here, but the gaps between them are small. That's the honest, common outcome - and it means the tie is broken by which forecast shape the decision needs, not by a hair of RMSE



What We're Really Grading Is a Trade

Every metric maps to a trading action

The scoreboard is not an academic exercise - each metric decides part of a trade. Accuracy sets how much to trust the point that fixes the hedge; the hit rate is the P&L of the direction; the log score grades the band that sizes the risk. We compare families to pick a trade, not a number

Metric -> trading job

- RMSE / MAE: trust in the point that sets the hedge.
- Hit rate: the direction that drives the P&L.
- Log score: the band that sizes the position.

So the winner is

- The family whose output the trade actually uses.
- Not the lowest decimal, but the best decision.
- Part 3 turns this straight into a pairs trade.

Small Gaps, Big Caveats

Don't over-read a thin margin

Why gaps are often small

- Financial returns are close to unpredictable.
- Much of the signal is common to all models.
- A tiny RMSE edge can be pure noise.

So be humble

- Check whether the gap survives another split.
- Prefer the simpler model when it's a near tie.
- Let the decision, not decimals, break the tie.

The Honest-Comparison Checklist

Six boxes to tick before you trust a result

Setup

- One asset, one clean train/test split.
- All models fit on train only.
- No statistic computed on the test.

Scoring

- Same metrics for every family.
- Every model compared to naive.
- The verdict tied to the decision.

Build the Comparison Harness

Direct Claude Code to grade the three families

GOAL Turn Part 1's three fitted families into a fair, out-of-sample scoreboard.

[INPUT] Prompt · Claude Code

Write and run `compare_families.py` that:

1. loads one asset and makes a train/test split (and, optionally, a walk-forward loop);
2. fits Classical, State-Space, and Bayesian models on TRAIN only - no statistic from the test;
3. forecasts the test period with each;
4. scores RMSE, MAE, hit rate, and a naive benchmark - plus a log score for the Bayesian;
5. prints a scoreboard and flags any tie.

Guard against look-ahead: fit scalars on train only.

How to work with it

- Same split for every model.
- Beat the naive bar first.
- Read several metrics, not one.
- Break ties by the decision.

Deliverable: `compare_families.py` + a scoreboard on identical metrics, guarded against look-ahead bias.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1 What makes a comparison fair?

**Same held-out test,
same metrics, no
leakage**

Q2 What is look-ahead bias?

**Using data the model
couldn't have had**

Q3 Why read several metrics together?

**A model can win one
and lose another**

All three come from the hold-out method, the look-ahead trap, and the finance metrics.

Three Pitfalls to Avoid

Where fair comparisons go wrong

Common mistakes

- "A great backtest means a great strategy."
- "The lowest RMSE always wins."
- "One train/test split is enough."

The reality

- Look-ahead bias fakes great backtests - guard against it.
- Read direction and calibration too, and mind small gaps.
- Walk forward so one lucky split can't decide it.

Three Things to Carry Forward

Part 2 in one screen

1

Fit on the past, grade on the future

Reserve a test period, fit every model on the earlier data only, and score each on the slice it never saw. Walk-forward averages over many honest forecasts instead of one lucky split.

2

Look-ahead bias is the cardinal sin

Any statistic or value the model couldn't have had at the time fakes great results that vanish live. Compute everything on train only, and never tune on the test.

3

Score like a desk, and read the fine print

Use size, direction, and calibration together, always against the naive bar. Gaps are often small - so let the decision, not a hair of RMSE, break the tie.

Next — Part 3: the final project end to end, from comparison to a trading decision, and the course close.

★ WEEK 16 · PART 2 · PRACTICE

The Fair Comparison, Worked by Hand

A Three-Way Scoreboard, Walk-Forward, and a Leak You Can See

Grade three forecasts; test twice; catch look-ahead with numbers

Part 2's method, with the arithmetic shown. We grade three forecasts on held-out days, watch the simplest win, then see how one split can flatter - and how a single leaked statistic inflates a score it shouldn't.

2P

PART 2 · APPLICATIONS

Two Comparisons, Computed by Hand

Case A: grade naive, AR, and a mean forecast on five held-out days and let RMSE decide. Case B: see one split flatter a model, and watch a leaked statistic inflate a z-score. Then a Claude Code lab.

What Part 2 Told Us

Now we make it concrete - with numbers

1

Fit on the past, grade on the future

Reserve a test period, fit on the earlier data only, and score each model on the slice it never saw.

2

Walk forward so one split can't fool you

A single split can flatter by luck. Averaging over many out-of-sample forecasts is more honest - we'll see the numbers move.

3

Look-ahead bias fakes great results

Any statistic computed with test data leaks the future. We compute an honest score and a leaked one side by side.

Everything here is a calculator exercise: five held-out days, two test setups, one leaked statistic.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a tiny, real-looking dataset.
- Compute the scores by hand, step by step.
- Then read what the numbers really say.

What you should watch

- How the simplest forecast can win.
- How one split flatters a model.
- How a leaked statistic inflates a score.

A Three-Way Scoreboard

Grade three forecasts, let RMSE decide

We forecast a return three ways: the naive forecast, an AR(0.4) model, and simply predicting the long-run mean. Which is best on this asset? We grade all three on five held-out days and compute their RMSE by hand.

The three candidates

- Naive: repeat the last return.
- AR(0.4): scale the last return by 0.4.
- Mean: just predict the average (here, zero).

What we'll do by hand

- Forecast five held-out days each way.
- Square the errors and sum them.
- Turn each sum into an RMSE and rank.

Three Ways to Forecast

From doing nothing to a model

The two simple ones

- Naive: tomorrow equals today's return.
- Mean: tomorrow equals the long-run average.
- Neither needs any fitting on features.

The model

- AR(0.4): tomorrow = 0.4 times today.
- It uses one estimated coefficient.
- The 'sophisticated' entry in the race.

The Held-Out Test: Five Days

Forecast each day three ways

Day	$r(t-1)$	actual	naive	AR(.4)	mean
1	+0.6	-0.2	0.6	0.24	0
2	-0.4	+0.5	-0.4	-0.16	0
3	+1.0	-0.6	1.0	0.40	0
4	-0.8	+0.7	-0.8	-0.32	0
5	+0.5	-0.3	0.5	0.20	0

Goal: score each forecast by its squared errors, then RMSE. naive = $r(t-1)$; AR = $0.4 \times r(t-1)$; mean = 0.

Score All Three

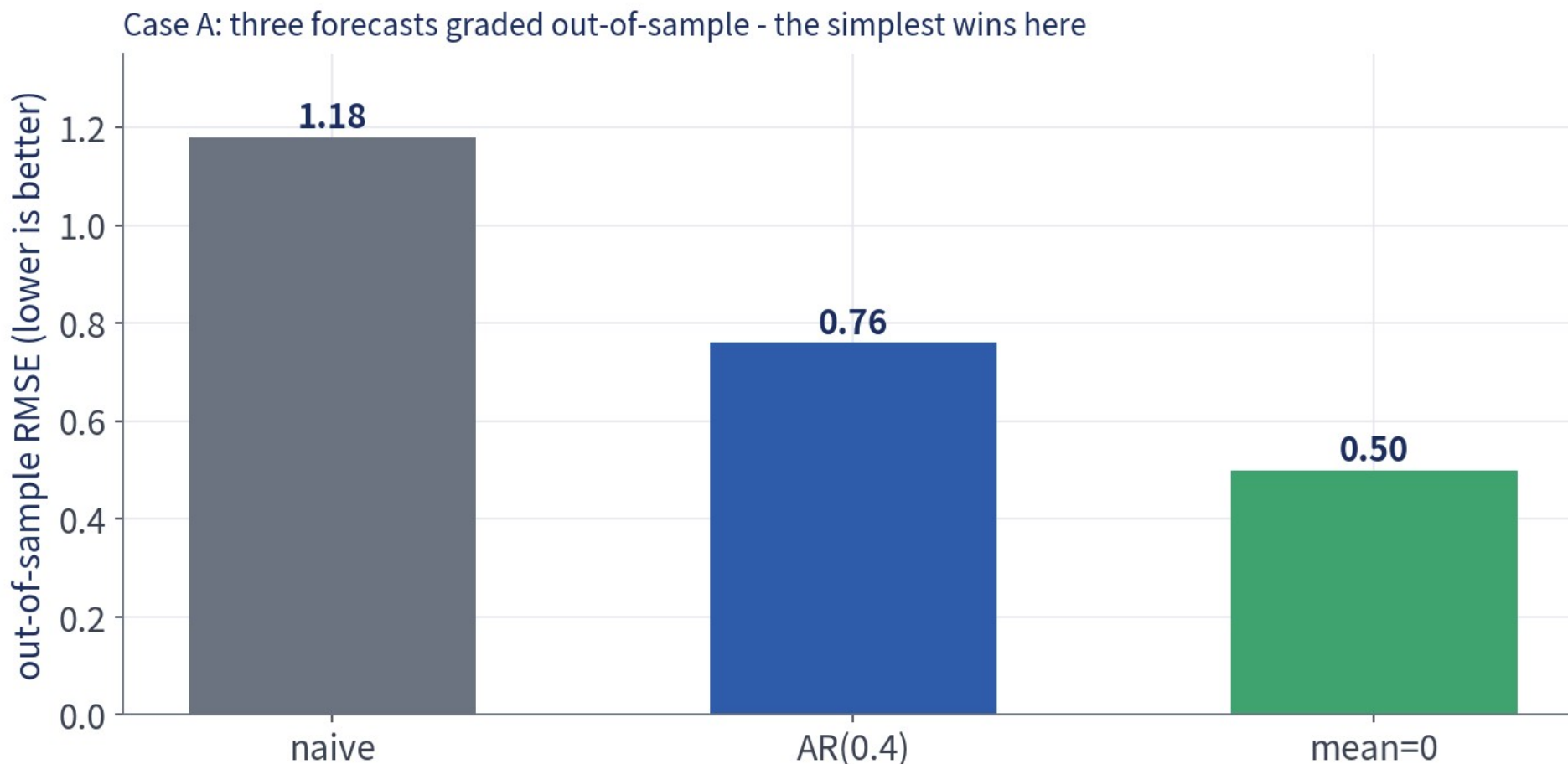
Squared-error sums to RMSE

naive: squared-error sum = 6.90 $\Rightarrow$ RMSE = $\sqrt{6.90/5} = 1.18$
AR(0.4) : squared-error sum = 2.92 $\Rightarrow$ RMSE = $\sqrt{2.92/5} = 0.76$
mean = 0 : squared-error sum = 1.23 $\Rightarrow$ RMSE = $\sqrt{1.23/5} = \mathbf{0.50}$
 $\Rightarrow$ the simplest forecast wins here - the opposite of Part 1

The mean forecast wins with RMSE 0.50 - the naive and the AR both do worse. Here, doing almost nothing is best.

The Three-Way Scoreboard

As a picture: the mean forecast has the lowest out-of-sample RMSE, the AR sits in the middle, and the naive is worst. On this asset the returns are close to unpredictable, so the plainest forecast wins.



Simple Wins Here

And that's the whole point

Compare to Part 1

- There, AR(1) beat the naive forecast.
- Here, the mean beats both.
- Different asset, different winner.

The lesson

- No forecast is best on every series.
- You cannot know the winner in advance.
- That is exactly why you always test.

Always Grade Against a Benchmark

Naive and mean are the bars

Why benchmarks

- The naive and mean forecasts cost nothing.
- For returns, they are often very hard to beat.
- They are the honest bar any model must clear.

The habit

- Report your model next to both benchmarks.
- If it can't beat them, prefer the simpler one.
- A model earns its keep only by scoring better.

One Split Can Lie - and Don't Peek

Two ways a test can mislead you

Two more traps, made numeric. First, a single train/test split can flatter a model just by luck; walk-forward averages that away. Second, a statistic computed with test data leaks the future and inflates the score. We compute both.

Trap 1: one lucky split

- A single split scores only a few points.
- It can land on an easy stretch.
- Walk-forward averages over many points.

Trap 2: a leaked statistic

- Standardizing with full-sample stats peeks.
- The 'test' data helped compute the scaler.
- The score is inflated, not earned.

Single Split vs Walk-Forward

Same model, two ways to grade it

Single split

- Fit once, forecast the last four days.
- Score those four out-of-sample errors.
- Fast - but only four points of evidence.

Walk-forward

- Forecast each day from the one before.
- Collect seven out-of-sample errors.
- More points, and a fairer average.

Grade the Same Model Twice

Single split vs walk-forward

single split (4 test points): $\text{RMSE} = 0.71$

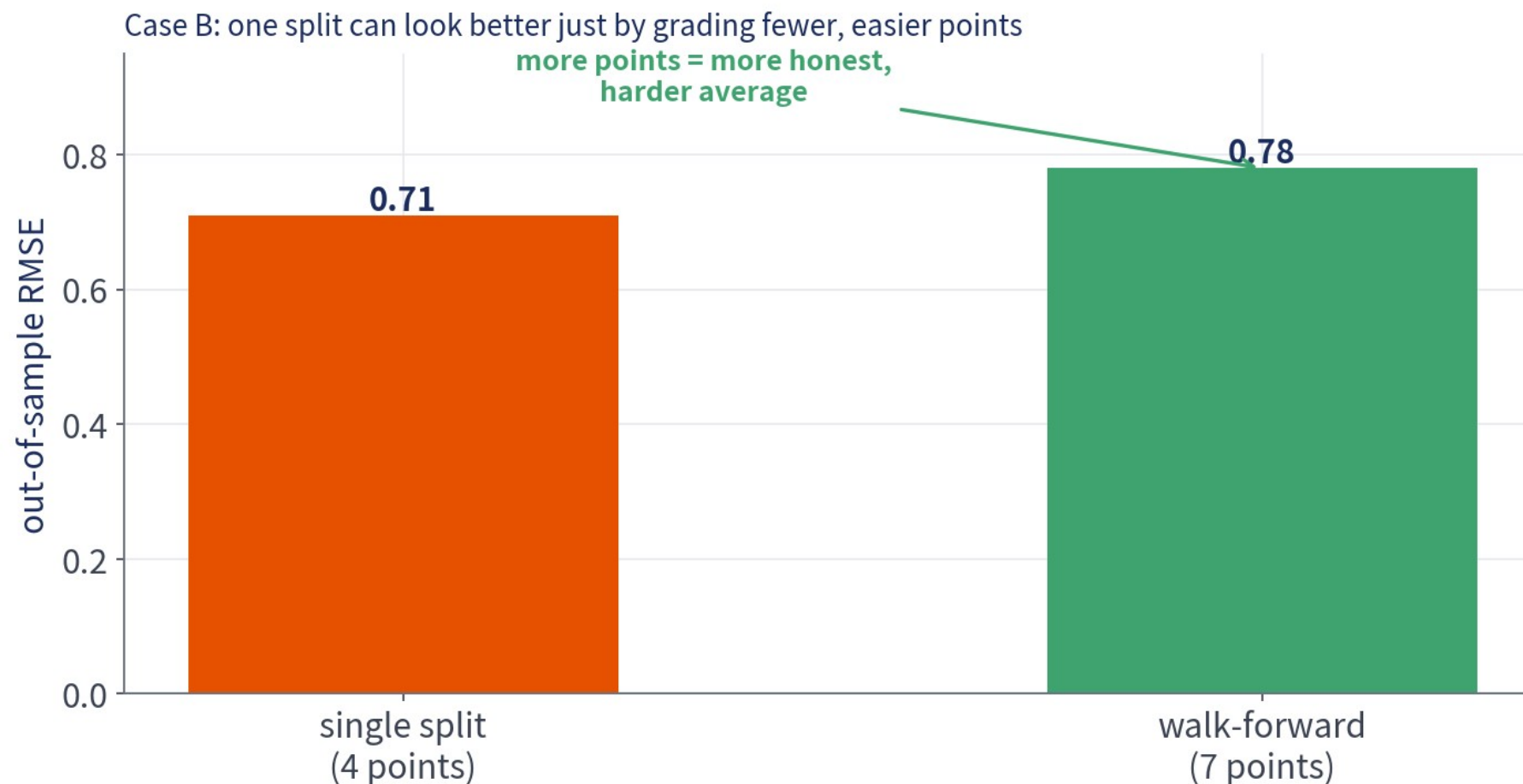
walk-forward (7 test points): $\text{RMSE} = 0.78$

⇒ one split looked better by grading fewer, easier points

The single split shows RMSE 0.71; walk-forward shows 0.78. The lower number wasn't skill - it just graded fewer, easier points.

One Split Flatters

The single split reports a lower RMSE than walk-forward for the very same model. It isn't better - it just graded four convenient points. Walk-forward's seven points give the more honest, and slightly higher, number.



Look-Ahead Bias, by the Numbers

One statistic can peek at the future

Look-ahead bias is subtle. Suppose you standardize returns into a z-score signal. If you use the standard deviation of the WHOLE series - including the test - your 'past' signal secretly knows the future. We compute the honest z and the leaked z and compare.

The honest way

- Compute the mean and sd on training only.
- Apply them to standardize the test points.
- Nothing from the future touches the scaler.

The leaked way

- Compute the sd over the whole series.
- That sd used the test data itself.
- Every z-score is quietly contaminated.

Honest z vs Leaked z

The same point, two standardizations

$$\begin{array}{ll} \text{train sd} = 0.66, & \text{full-sample sd} = 0.56 \text{ (secretly uses the test)} \\ \text{honest: } z = \frac{0.40 - 0.10}{0.66} = 0.46, & \text{leaked: } z = \frac{0.40 - 0.10}{0.56} = \mathbf{0.54} \end{array}$$

The leaked z (0.54) is larger than the honest one (0.46) because its denominator used the test data - a future it should never have seen.

What Actually Leaked

A small number, a big lesson

It looks harmless

- The two z-scores differ only a little.
- But the leaked one used the test to scale.
- Scaled across a whole backtest, it compounds.

Why it's dangerous

- Leaked signals look sharper than they are.
- The edge evaporates in live trading.
- Compute every statistic on training only.

The Common Pattern

Two comparisons, one discipline

Both cases, computed by hand, protect you from fooling yourself. Grade every forecast against simple benchmarks and let the numbers rank them. Average over many out-of-sample points, and compute every statistic on training data only.

The traps

- Assuming the fancy model wins.
- Trusting one lucky split.
- Leaking a statistic from the test.

The fixes

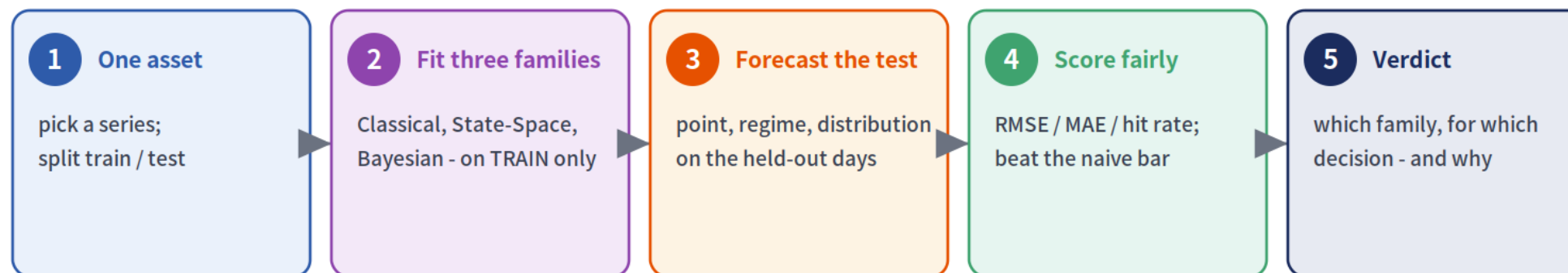
- Grade against naive and mean.
- Walk forward over many points.
- Fit every scaler on train only.

Where the Hand Calcs Fit

Case A is the 'score fairly' box; Case B guards the two honesty rules - many out-of-sample points, and no statistic from the test. Together they turn the capstone pipeline from a diagram into arithmetic you can check

The Capstone as a Pipeline

One asset in, three fitted families, one held-out grade, one honest verdict out.



The two rules that keep it honest

1. Fit on the past only - never let a model touch the test data before it is graded (no look-ahead bias).
2. Compare every family on the SAME held-out days, against the naive benchmark, with the same metrics.

The verdict is decision-driven: the winner is the family whose output your decision actually needs.

Reproduce the Hand Calcs, Then Scale Up

Direct Claude Code to check your arithmetic

GOAL Verify both comparisons in code, then extend them to a real asset.

[INPUT] Prompt · Claude Code

Write and run `part2_practice.py` that:

CASE A - scoreboard:

- * take the 5-day table; form naive, AR(0.4), and mean forecasts; print squared-error sums;
- * compute and rank the three RMSEs.

CASE B - honesty:

- * grade one AR model by a single split and by walk-forward; print both RMSEs;
- * standardize a test point with train-only sd and with full-sample sd; print both z-scores.

Match every printed number to the slides, then repeat on a real return series.

Check against the slides

- RMSE naive 1.18, AR 0.76, mean 0.50.
- Single split 0.71, walk-forward 0.78.
- Honest z 0.46, leaked z 0.54.
- Simple wins; one split flatters; leaks inflate.

Deliverable: `part2_practice.py` whose printed numbers match the hand calculations, then a real asset.

Three Questions on the Cases

If you can answer these, the practice has landed

Q1 Why did the mean forecast win Case A?

Lowest out-of-sample RMSE on this asset

Q2 Why was the single-split RMSE lower?

It graded fewer, easier points

Q3 Why was the leaked z-score larger?

Its sd used the test data

All three come from the scoreboard, the walk-forward comparison, and the leaked standardization.

Three Pitfalls to Avoid

Where comparisons mislead

Common mistakes

- "The model must beat the naive forecast."
- "One good split proves the model."
- "Standardizing on all the data is fine."

The reality

- Sometimes the mean wins - so always test.
- One split can flatter; walk forward.
- Full-sample stats leak the future - use train only.

Two Comparisons to Remember

Part 2 applications in one screen

1

Grade three forecasts; let the number decide

Naive, AR(0.4), and a mean forecast scored RMSE 1.18, 0.76, and 0.50. The simplest won on this asset - the reverse of Part 1 - which is exactly why you never assume and always test against a benchmark.

2

One split can flatter; walk forward

The same model scored 0.71 on a single split but 0.78 walk-forward. The lower number wasn't skill; it just graded fewer, easier points. Many out-of-sample points give the honest average.

3

A single leaked statistic inflates the score

Standardizing with a full-sample sd (0.56) instead of the train sd (0.66) turned an honest z of 0.46 into a leaked 0.54. Compute every statistic on training data only.

Next — Part 3: the final project end to end, from comparison to a trading decision, and the course close.

★ WEEK 16 · PART 3 of 3

The Final Project, and the Course Close

From an Honest Comparison to a Trading Decision

The project, step by step · the competencies you carry · what's next

The finale. We turn the fair comparison into a deliverable: one asset, three families, a held-out test, and a verdict that becomes a trading recommendation. Then we close the core course and look ahead.

3

PART 3 · THE FINAL PROJECT

Comparison to Decision to Delivery

Your final project step by step, turning the winning model into a trading decision, presenting the evidence honestly, the three competencies you now hold, and where the journey goes next.

From Method to Deliverable

Parts 1-2 built it; Part 3 ships it

1

Part 1 gave us the choice

Three families, three forecast shapes - matched to the decision at hand.

2

Part 2 gave us the test

A fair, out-of-sample comparison against the naive benchmark, guarded against look-ahead bias.

3

Part 3 gives us the deliverable

One asset run end to end, an honest verdict, and a trading recommendation you can defend.

Part 3 in one line: turn the comparison into a decision, present it honestly, and close the course.

Your Final Project, Step by Step

One asset, three families, a fair test, a verdict

The capstone is a complete, honest study of one asset. You pick the series, split it cleanly, fit all three families on the past, grade them on the future, and end with a verdict that ties the winner to a real trading decision - caveats included.

What you produce

- A clean train/test (or walk-forward) setup.
- A forecast from each of the three families.
- A scoreboard and a decision-driven verdict.

What makes it good

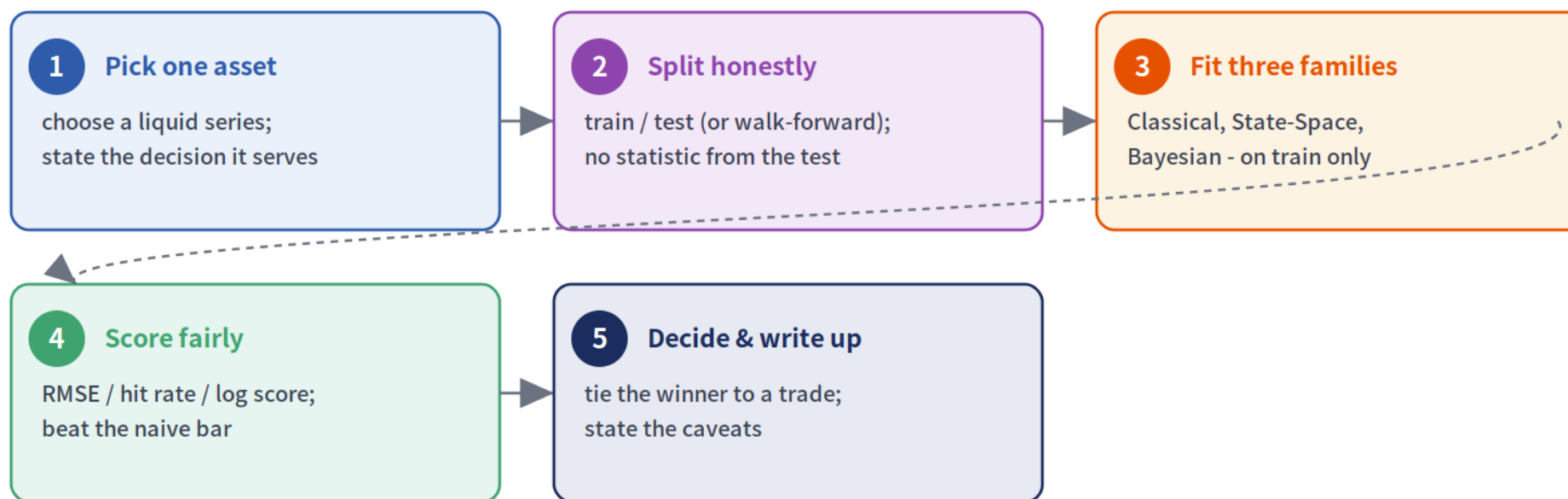
- Honesty: no leakage, real out-of-sample scores.
- Clarity: the decision the study serves is stated.
- Humility: the caveats are as visible as the result.

Your Final Project, Step by Step

Pick one asset and state the decision; split honestly; fit all three families on the training data; score them fairly against the naive bar; then decide and write it up. The deliverable is the verdict, not the model!

Your Final Project, Step by Step

One asset, three families, a fair test, an honest verdict - and a trading recommendation.



The deliverable is not the model - it is the honest verdict and the decision it supports.

Pick the Asset and the Decision

Start from what you're trying to do

Choosing the asset

- A liquid series with enough history.
- Something whose behavior you can reason about.
- The pair from your thread is a natural choice.

Stating the decision

- What will the forecast actually support?
- A trade signal, a hedge, or a risk limit?
- The decision sets which forecast shape wins.

Split Honestly, Fit Three Families

The disciplined middle of the project

Split honestly

- Reserve the most recent slice as the test.
- Consider walk-forward for a stronger grade.
- Compute no statistic on the test data.

Fit three families

- Classical, state-space, and Bayesian.
- All on the training period only.
- One clear representative from each.

Score Fairly, Then Decide

Turn the scoreboard into a verdict

Score fairly

- RMSE, hit rate, and a log score for the band.
- Every family against the naive benchmark.
- Read several metrics together, not one.

Then decide

- Pick the family whose output the decision needs.
- Break near-ties with simplicity.
- State the verdict in one honest sentence.

From Comparison to a Trading Decision

Tie the models back to the pairs strategy

A scoreboard is not the finish line - a decision is. The winning forecast feeds the pairs strategy: a point sets the hedge, a regime probability gates the trade, and a distribution sizes the risk. The final test is whether it improves the out-of-sample P&L.

How each output is used

- A point forecast sets the hedge ratio.
- A regime probability gates on/off and size.
- A distribution sets the risk and the stop.

The real verdict

- Does the signal beat buy-and-hold out of sample?
- Is the improvement worth the complexity?
- Would you actually trade it?

One Pair Trade, Every Model

This is where all the time series lands. Cointegration sets the hedge ratio and the spread; the z-score triggers the entry; OU gives the holding period; GARCH sizes it; the HMM gates it; and the Bayesian band sets the conviction. Six models, one trade ticket.

One Pair Trade, Every Model

The whole course, assembled into a single decision - each model contributes one number.



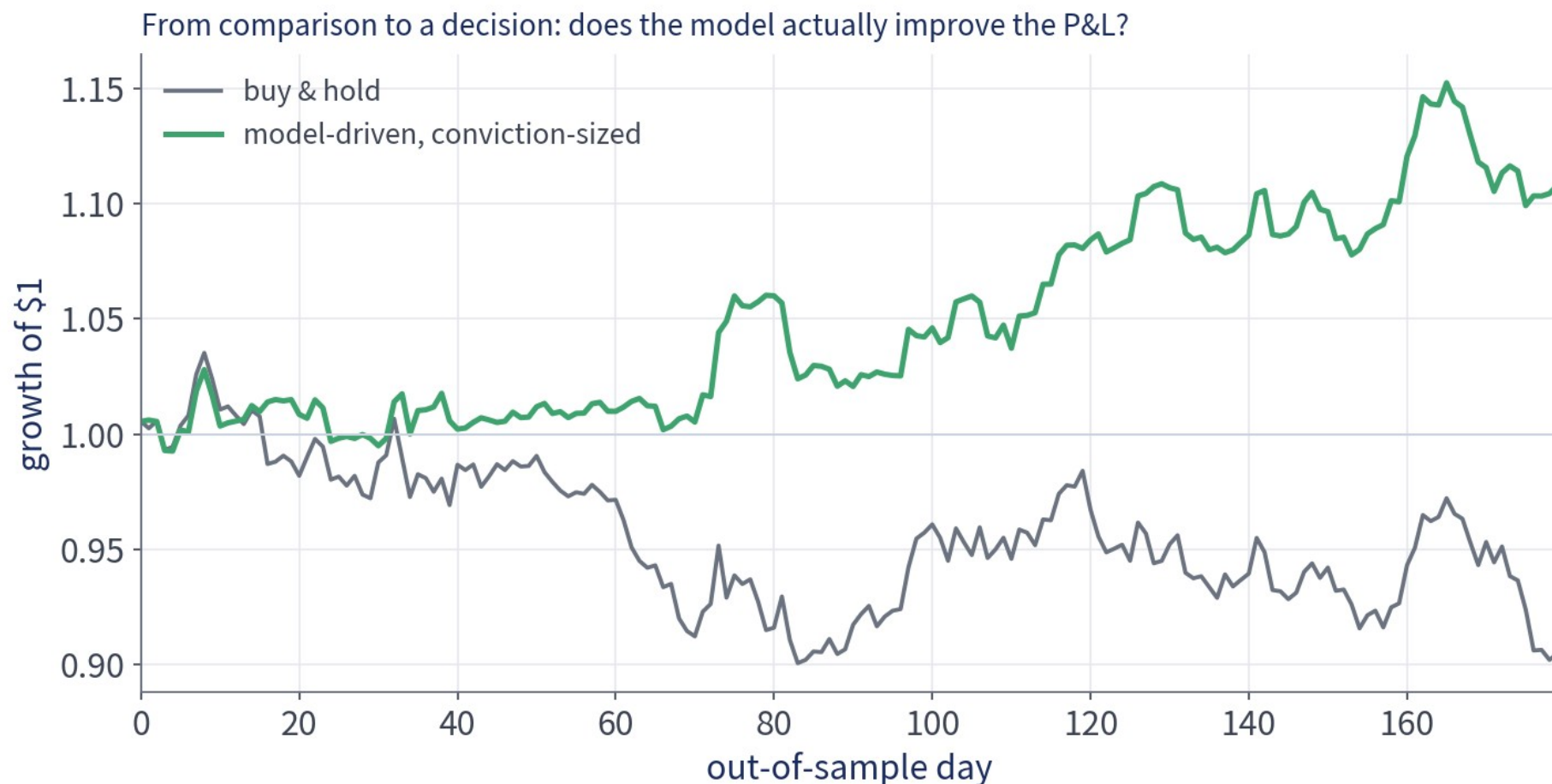
The trade ticket

SHORT the spread (short A, long 0.76 B), size 0.32 of the book, expected exit near $z = 0$ in about 2 days, stop out if P(crisis) rises above 0.5. Every number came from a model you built this course.

That is what all the time series was for.

Does the Model Improve the P&L?

The ultimate grade for a trading study: the model-driven, conviction-sized strategy against buy-and-hold, out of sample. A lower error only matters if it turns into a better, and honestly tested, equity curve.



Presenting It - and Pitfalls to Avoid

Make the evidence clear and the honesty obvious

A good capstone is judged as much on honesty as on results. Lead with the decision, show the fair test, and be upfront about the caveats. The most impressive thing you can present is a result you have genuinely tried to break.

How to present

- State the decision and the asset first.
- Show the split, the metrics, and the naive bar.
- End with a clear, one-line recommendation.

Pitfalls to avoid

- Hiding the caveats or the failed attempts.
- Over-claiming from a thin RMSE gap.
- A backtest you haven't stress-tested for leakage.

Three Competencies You Now Have

What you take out of the core course

Beyond any single model, you leave with three durable skills. You can match a model to a decision, you can test any claim honestly out of sample, and you can let uncertainty size the bet. These outlast any particular technique.

The three skills

- Match the model to the decision.
- Test every claim out of sample.
- Let honest uncertainty size the bet.

Why they last

- They apply to models we haven't met yet.
- They are how professionals avoid fooling themselves.
- They turn techniques into judgment.

Match the Model to the Decision

Start from the choice, not the tool

What it means

- Ask what the decision needs first.
- Pick the forecast shape that fits.
- Point, regime, or distribution.

Why it matters

- It stops you reaching for the fanciest tool.
- It keeps the work tied to a real purpose.
- It makes 'which model?' a solvable question.

Test Everything Out of Sample

The habit that keeps you honest

What it means

- Fit on the past, grade on the unseen future.
- Compare against a naive benchmark.
- Guard fiercely against look-ahead bias.

Why it matters

- It is how the field settled its arguments.
- It separates real edges from lucky fits.
- It is the difference between a demo and a strategy.

Let Uncertainty Size the Bet

The Bayesian habit, made practical

What it means

- Carry a credible band, not just a point.
- Scale exposure by conviction and by the band.
- Trade smaller when the model is unsure.

Why it matters

- It is the lesson of the quant quake.
- It turns honesty about doubt into safer risk.
- It connects the whole course to real trading.

What's Next: the Deep-Learning Special

Four bonus weeks - and the skepticism to keep

The core course ends here, but the journey continues. A deep-learning special covers neural forecasters and sequence models - powerful tools that deserve exactly the same scrutiny. Bring your out-of-sample discipline with you; it matters more, not less, as models grow

What's ahead

- Neural networks for time series.
- Sequence models and attention.
- When deep learning helps - and when it doesn't.

What to bring

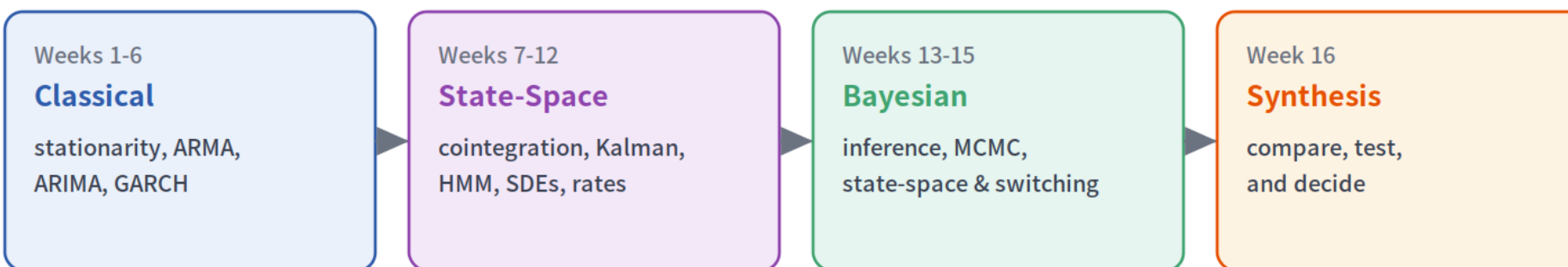
- The same fair, out-of-sample tests.
- The same naive benchmark.
- The same honesty about uncertainty.

The Core Course, Complete

Sixteen weeks across three families, one pairs thread running through them, and a deep-learning special ahead - to be held to the very same out-of-sample bar. Carry three things forward: decision, testing and honest uncertainty

The Core Course, Complete

Sixteen weeks, three families, one pairs thread - and the skepticism to carry forward.



What's next: the Deep-Learning special (four bonus weeks)

Neural forecasters and sequence models - powerful, and to be held to the very same out-of-sample bar.

Carry three things forward

Match the model to the decision; test everything out-of-sample; and let honest uncertainty size the bet.

Three Families on One Asset

Assemble the core of your capstone

GOAL Produce the full comparison for one asset, end to end, ready to write up.

[INPUT] Prompt · Claude Code

Write and run `capstone_compare.py` that:

1. loads one asset and makes an honest split;
 2. fits Classical, State-Space, and Bayesian models on the training data only;
 3. forecasts the test period with each family;
 4. scores RMSE, hit rate, and a log score against the naive benchmark, and prints a scoreboard;
 5. saves the forecasts and scores for the write-up.
- Flag any look-ahead risk it finds in the pipeline.

How to work with it

- One asset, one honest split.
- Three families, same test.
- Beat the naive bar.
- Keep the scoreboard for the report.

Deliverable: `capstone_compare.py` + a saved scoreboard and forecasts for one asset.

From Forecasts to a Trading Verdict

Turn the comparison into a recommendation

GOAL Convert the winning forecast into a trading decision and grade its out-of-sample P&L.

[INPUT] Prompt · Claude Code

Write and run `capstone_decision.py` that:

1. takes the winning family's forecast;
 2. turns it into a pairs-trade rule - a hedge from a point, a gate from a regime, a size from a band;
 3. simulates the strategy on the TEST period only;
 4. compares its equity curve and Sharpe to buy-and-hold;
 5. writes a one-paragraph verdict with caveats.
- State plainly whether you would trade it.

How to work with it

- Use only the test period.
- Compare to buy-and-hold.
- Report Sharpe, not just error.
- End with a clear recommendation.

Deliverable: `capstone_decision.py` + an out-of-sample equity curve and a written, caveated verdict.

Three Questions Before You Finish

If you can answer these, the course has landed

Q1 What is the real deliverable of the capstone?

**An honest verdict
and a decision**

Q2 What is the ultimate test of a trading study?

**Out-of-sample P&L vs
buy-and-hold**

Q3 What three competencies do you carry forward?

**Decision, testing,
honest uncertainty**

All three come from the project steps, the P&L verdict, and the three durable skills.

Three Pitfalls to Avoid

Where capstones go wrong

Common mistakes

- "A great scoreboard is the finished project."
- "Present only the version that worked."
- "The fanciest model must be the answer."

The reality

- The deliverable is the decision, not the score.
- Show the caveats and the failures too.
- Let the decision and the honest test decide.

Three Things to Remember

The whole week in one screen

1

Match the model to the decision

Three families answer three questions - point, regime, distribution. Start from what the decision needs, and there is no universal best model.

2

Test everything out of sample

Fit on the past, grade on the unseen future, beat the naive benchmark, and guard against look-ahead bias. The scoreboard, not taste, decides.

3

Deliver a decision, honestly

Turn the winning forecast into a trading recommendation, grade its out-of-sample P&L, and present the caveats as clearly as the result.

The core course is complete - three families, one thread, and the discipline to use them well.

Where You Go From Here

The core course is complete

You started with a single return series and a question about stationarity. You end able to model its mean and variance, track its hidden states, quantify its uncertainty, and test any claim about it honestly. That arc - from data to decision - is the whole point.

What you can now do

- Model returns with the classical family.
- Track hidden states and regimes.
- Quantify uncertainty the Bayesian way.

And crucially

- Choose among them for a real decision.
- Test every claim out of sample.
- Size the bet to honest uncertainty.

Keep the Skepticism

The habit worth more than any model

As models get fancier

- The temptation to trust them grows.
- So does the risk of fooling yourself.
- Deep learning is no exception.

So keep asking

- What decision does this serve?
- How does it do out of sample?
- How sure am I, really?

Three Families, One Thread, One Discipline

Advanced Financial Time Series

1

You learned three families

Classical for the mean and variance, state-space for hidden states and regimes, and Bayesian for honest, full uncertainty.

2

You followed one thread

The pairs strategy grew across sixteen weeks, from finding a pair to trading it with credible bands.

3

You keep one discipline

Match the model to the decision, test everything out of sample, and let honest uncertainty size the bet.

Thank you - and good luck with the final project.

★ WEEK 16 · PART 3 · PRACTICE

From Comparison to a Trade, by Hand

Grade a Strategy's P&L; Size the Bet by Conviction

A signal vs buy & hold - and a bet that shrinks when unsure

Part 3's finale, with the arithmetic shown. We turn a winning forecast into a trading rule, grade its out-of-sample P&L against buy-and-hold, then size the final bet by conviction and the width of the credible band.

3P

PART 3 · APPLICATIONS

Two Steps, Computed by Hand

Case A: turn a signal into a strategy, grade its P&L and Sharpe on six held-out days against buy-and-hold. Case B: size the final bet by edge, conviction, and band width. Then a Claude Code lab.

What Part 3 Told Us

Now we make it concrete - with numbers

1

The deliverable is a decision

A scoreboard is not the finish line. The winning forecast has to become a trade you can defend.

2

The real test is out-of-sample P&L

Does the strategy beat buy-and-hold on data it never saw? We compute the P&L, the Sharpe, and the hit rate by hand.

3

Uncertainty sizes the final bet

Edge, conviction, and the width of the credible band together set how much to commit. We compute the size by hand.

Everything here is a calculator exercise: six held-out days, one signal, one sizing rule.

How These Cases Work

Data, then arithmetic, then meaning

Each case has three steps

- Start from a signal or a set of inputs.
- Compute the P&L or the size by hand.
- Then read the trading decision it supports.

What you should watch

- Whether the strategy actually beats buy & hold.
- How the bet shrinks as the band widens.
- How the whole course lands on one trade.

From a Forecast to a Verdict

Does the signal actually make money?

The comparison picked a winning forecast; now it must earn its keep. We turn its directional signal into a strategy - go with the sign of the forecast - and grade its P&L over six held-out days against simply buying and holding.

The strategy rule

- Take the sign of the model's forecast.
- The strategy return is sign times the market.
- A right call captures the move; a wrong one loses it.

What we'll do by hand

- Compute the strategy return each day.
- Sum the P&L and compare to buy & hold.
- Compute the Sharpe and the hit rate.

Turning a Signal Into a Trade

Sign of the forecast times the market

How it works

- Forecast up: go long, earn the market return.
- Forecast down: go short, earn its negative.
- So strategy return = $\text{sign}(\text{signal}) \times \text{market}$.

Why grade it this way

- It is the simplest use of a directional call.
- It reveals whether the signal has real edge.
- The benchmark is always buy-and-hold.

The Held-Out Test: Six Days

Signal, market, and the strategy return

Day	signal	market (%)	strategy = sign x market
1	+ (up)	+0.8	+0.8
2	+ (up)	-0.5	-0.5
3	- (down)	-1.2	+1.2
4	+ (up)	+0.6	+0.6
5	- (down)	+0.4	-0.4
6	+ (up)	+1.0	+1.0

Goal: sum the strategy column, compare to summing the market column, then compute Sharpe and hit rate.

P&L, Sharpe, and Hit Rate

Grade the strategy against buy & hold

$$\text{strategy}_t = \text{sign}(\text{signal}) \times \text{market}_t$$

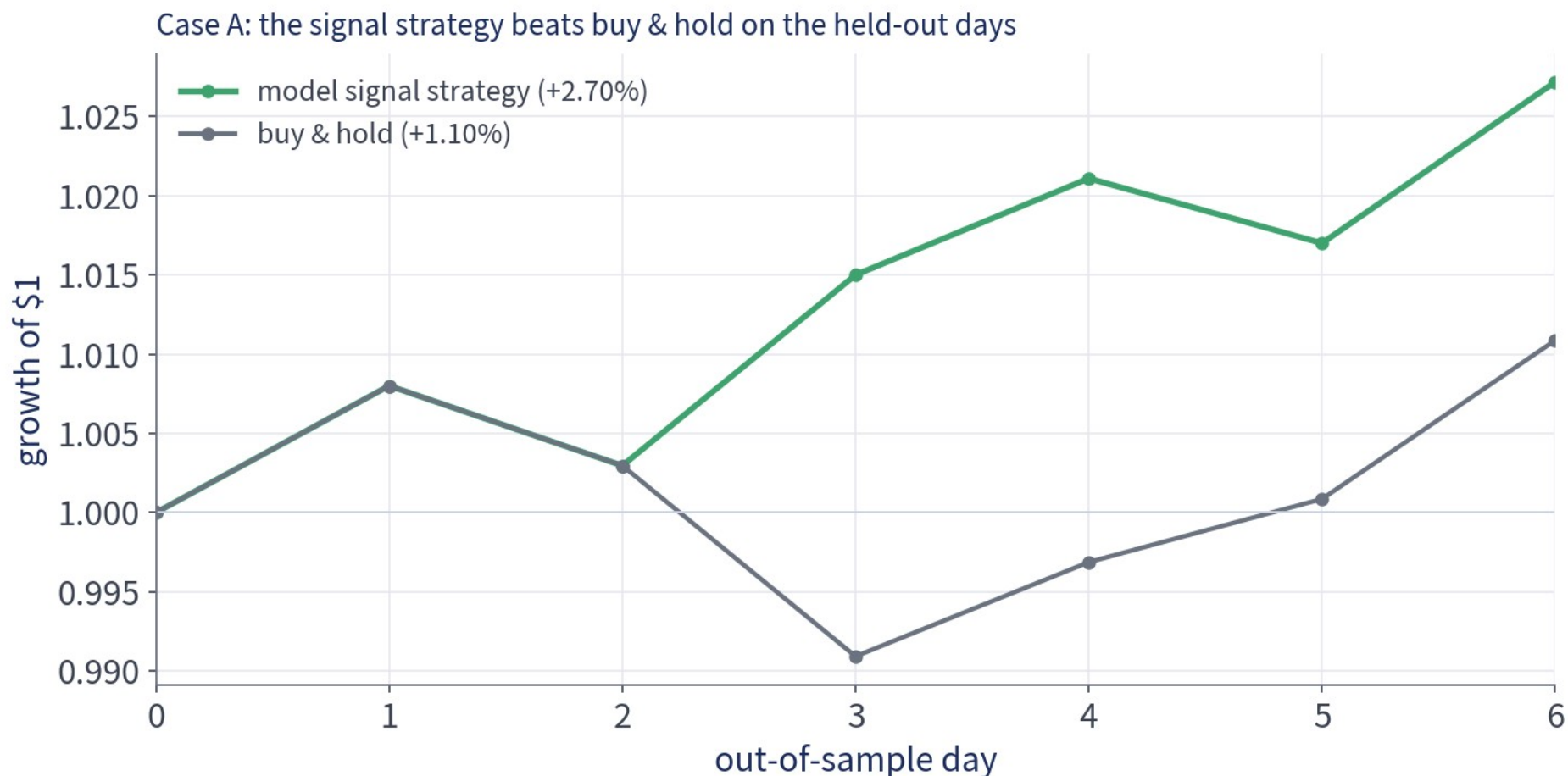
total P&L: strategy = +2.70%, buy & hold = +1.10%

Sharpe (per day): $\frac{0.45}{0.73} = \mathbf{0.62}$ vs $\frac{0.18}{0.85} = 0.21$, hit rate = $\frac{4}{6} = 67\%$

The strategy makes +2.70% versus +1.10% for buy & hold, with a higher per-day Sharpe and a 4-of-6 hit rate.

The Strategy Beats Buy & Hold

The same numbers as an equity curve. On the held-out days the signal strategy ends higher than buy-and-hold - the ultimate grade for a trading study. But six days is short, so treat it as an illustration, not proof.



The Real Verdict

Beating the benchmark, honestly

What to ask

- Did it beat buy-and-hold out of sample?
- Is the edge worth the extra complexity?
- Does it survive a longer test and costs?

What to resist

- Reading too much into six days.
- Ignoring trading costs and slippage.
- Claiming victory on a thin margin.

Size the Bet by Conviction

The final decision the whole course points to

A verdict is not enough - you must decide how much to commit. Conviction sizing multiplies the edge by the probability the setup is favorable and by how tight the credible band is. When the band is wide, the bet shrinks. We compute it by hand.

The three inputs

- Edge: the expected move if right.
- Conviction: $P(\text{favorable})$, from the regime.
- Band: how tight the forecast is.

What we'll do by hand

- Compute the base conviction size.
- Adjust it for a tight and a wide band.
- See the bet shrink as uncertainty rises.

Sizing by Conviction

Edge, conviction, and the band

The logic

- Start from the raw edge.
- Scale by how sure the regime is favorable.
- Shrink when the credible band is wide.

The behavior

- Full size only when all three agree.
- Automatic caution when the model is unsure.
- Uncertainty sets the bet, not a footnote.

Compute the Bet Size

Edge x conviction x band

$$\text{size} = \text{edge} \times P(\text{favorable}) \times (1 - \text{band})$$

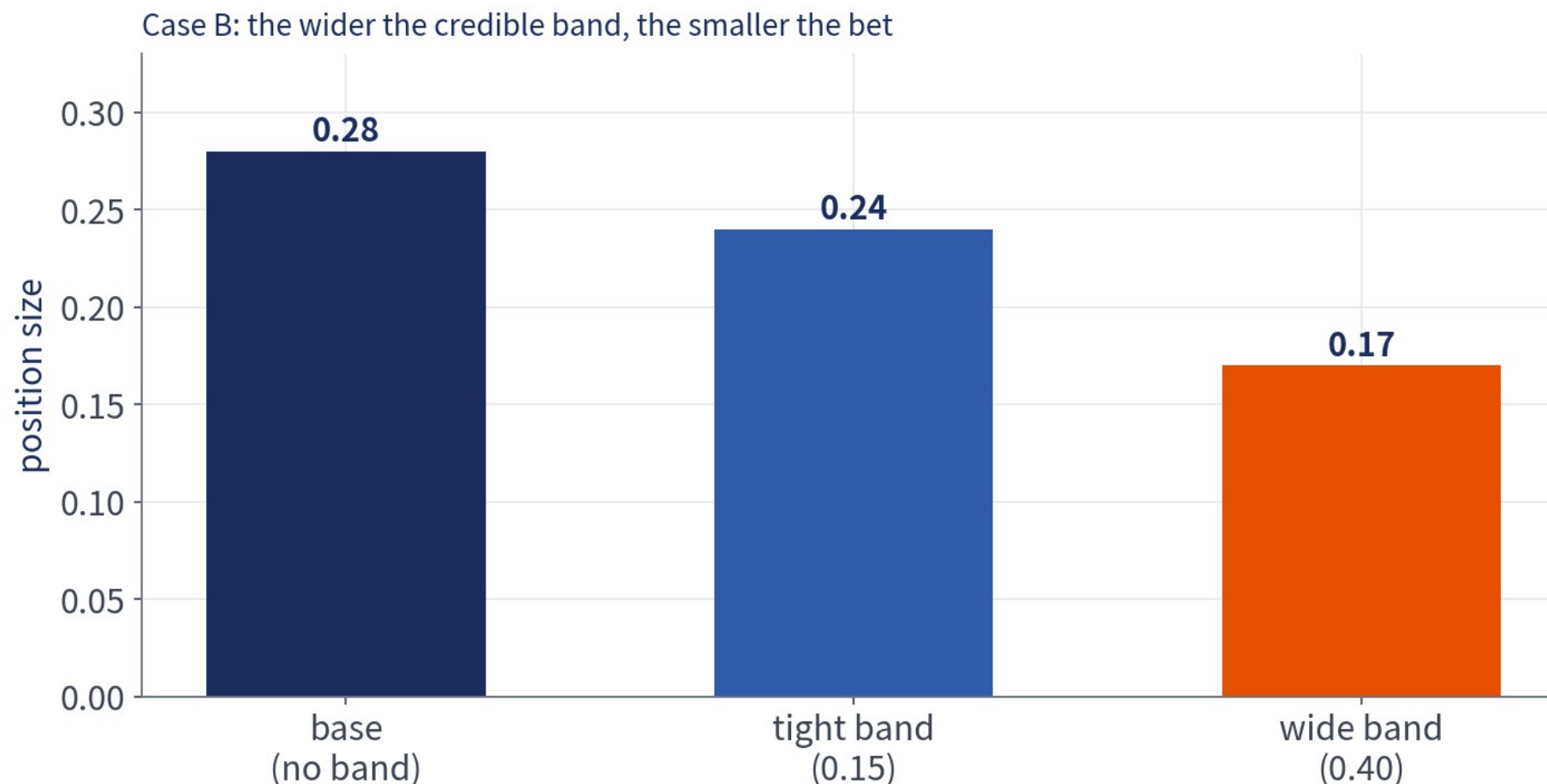
$$\text{base} = 0.40 \times 0.70 = 0.28$$

$$\text{tight band (0.15)} : 0.28 \times 0.85 = 0.24, \quad \text{wide band (0.40)} : 0.28 \times 0.60 = \mathbf{0.17}$$

With a tight band the bet is 0.24; with a wide band it falls to 0.17 - about 30% smaller, purely because the model is less sure.

A Wider Band, a Smaller Bet

The base conviction size shrinks step by step as the credible band widens. The model's honesty about its own uncertainty flows directly into how much capital you put at risk - exactly the quant-quake lesson.



What Conviction Sizing Buys

A little upside for much less regret

The trade-off

- A smaller bet gives up some expected gain.
- But it loses far less when the call is wrong.
- You commit most only when you are most sure.

Why it's worth it

- It caps the damage from an overconfident model.
- It keeps you in the game through bad stretches.
- It turns honest doubt into safer risk.

One Pair Trade, Every Model

The whole course, assembled by hand

Here is the capstone in one worked example: a single pairs trade in which every time-series model you built contributes one number. Cointegration, a z-score, OU, GARCH, an HMM, and a Bayesian band together produce a complete trade ticket - computed by hand.

Each model's job

- Cointegration: the hedge ratio and the spread.
- z-score & OU: the entry and the holding period.
- GARCH, HMM, Bayesian: the size and the gate.

The output

- A direction: short or long the spread.
- A size: the fraction of the book to risk.
- An exit and a stop, from the models.

The Spread and the Entry

Cointegration sets the pair; the z-score triggers

$$(1) \text{ Cointegration: } \beta = \frac{\text{Cov}(A, B)}{\text{Var}(B)} = \frac{0.070}{0.092} = 0.76, \quad \text{spread} = A - 0.76 B$$

$$(2) \text{ z-score today: } z = \frac{\text{spread} - \mu}{\sigma} = +2.1 \Rightarrow \text{spread is rich, SHORT it}$$

Cointegration (W7) gives the hedge ratio 0.76 and the spread; today's z-score of +2.1 says the spread is rich - so we SHORT it.

The Holding Period

OU turns mean-reversion speed into days

$$(3) \text{ OU half-life} = \frac{\ln 2}{-\ln \phi} = \frac{0.693}{-\ln 0.75} = \frac{0.693}{0.288} = \mathbf{2.4} \text{ days (holding period)}$$

What it gives

- The spread's AR(1) coefficient is 0.75.
- Its half-life is about 2.4 days.
- So we expect to hold the trade ~2 days.

Why it matters

- It sets a realistic exit horizon.
- It tells you if the edge decays too slowly.
- OU (Week 11) made this a number.

The Size and the Gate

GARCH sizes, the HMM gates, Bayes sets conviction

$$(4) \text{ GARCH size: } \frac{\text{target risk}}{\text{vol}} = \frac{0.010}{0.018} = 0.56$$

$$(5) \text{ HMM gate: } P(\text{crisis}) = 0.30 \Rightarrow \text{gate} = 1 \quad (6) \text{ conviction } 0.72, \text{ band } 0.20$$
$$\text{final size} = 0.56 \times 0.72 \times (1 - 0.20) \times 1 = \mathbf{0.32}$$

GARCH (W5-6) vol-targets to 0.56; the HMM (W10) gate is ON since $P(\text{crisis})$ is low; the Bayesian band and conviction (W15) trim it to a final size of 0.32.

One Pair Trade, Every Model

Assembled: SHORT the spread (short A, long 0.76 B), size 0.32 of the book, exit near $z = 0$ in about two days, stop out if $P(\text{crisis})$ rises above 0.5. Every number on the ticket came from a model you built.

That is what all the time series was for

One Pair Trade, Every Model

The whole course, assembled into a single decision - each model contributes one number.



The trade ticket

SHORT the spread (short A, long 0.76 B), size 0.32 of the book, expected exit near $z = 0$ in about 2 days, stop out if $P(\text{crisis})$ rises above 0.5. Every number came from a model you built this course.

That is what all the time series was for.

The Common Pattern

Two calculations, one deliverable

Both cases, computed by hand, turn analysis into action. A comparison becomes a strategy graded honestly on its P&L. A forecast becomes a bet whose size reflects real conviction. Together they are the capstone: an honest verdict, and a trade you can defend

The trap

- Stopping at a scoreboard.
- Betting the same regardless of conviction.
- Claiming an edge you never traded out of sample.

The fix

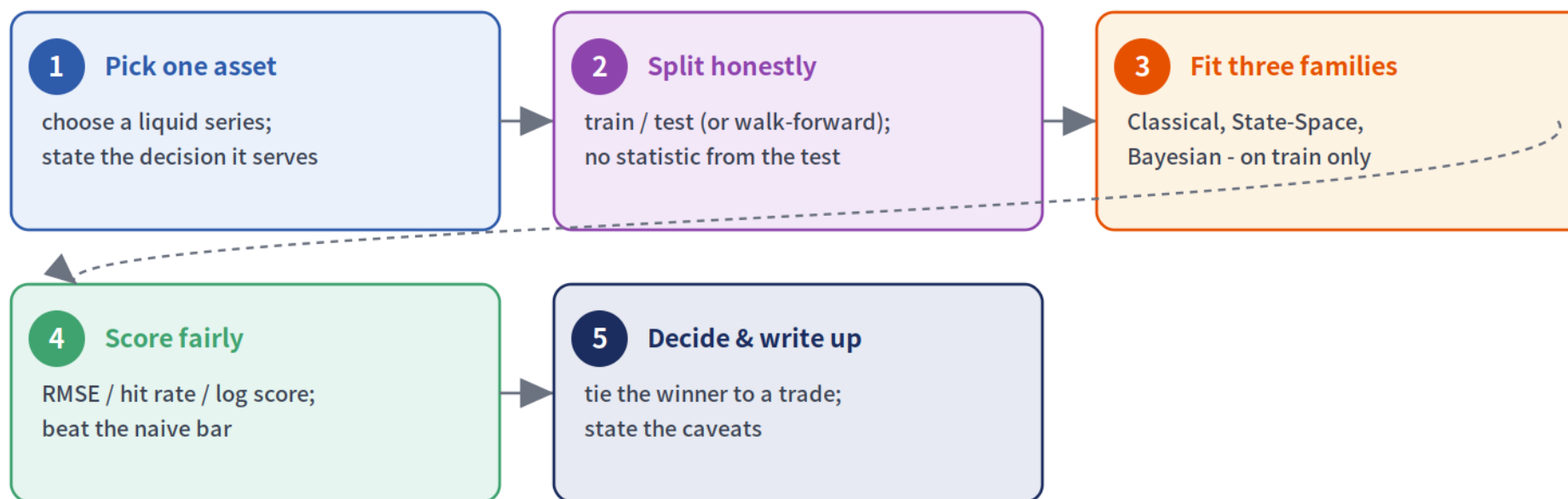
- Grade the strategy's P&L against buy & hold.
- Size the bet by edge, conviction, and band.
- Deliver a decision, with its caveats.

Where the Hand Calcs Fit

Case A is the 'decide and write up' box - grading the strategy's P&L. Case B is the sizing that turns the verdict into a bet. Together they complete the final project pipeline, from one asset to one defensible trade.

Your Final Project, Step by Step

One asset, three families, a fair test, an honest verdict - and a trading recommendation.



The deliverable is not the model - it is the honest verdict and the decision it supports.

Reproduce the Hand Calcs, Then Backtest

Direct Claude Code to check your arithmetic

GOAL Verify both steps in code, then run the full strategy on a real asset.

[INPUT] Prompt · Claude Code

Write and run `part3_practice.py` that:

CASE A - grade the strategy:

- * take the 6-day signal/market table; compute strategy = sign x market; print daily P&L;
- * print total P&L, per-day Sharpe, and hit rate for strategy and buy-and-hold.

CASE B - size the bet:

- * compute size = edge x P(favorable) x (1-band);
- * print the base, tight-band, and wide-band sizes.

Then run the signal strategy on a real asset, out of sample, with costs, vs buy-and-hold.

Check against the slides

- Strategy +2.70% vs buy-and-hold +1.10%.
- Per-day Sharpe 0.62 vs 0.21.
- Hit rate $4/6 = 67\%$.
- Size: base 0.28, tight 0.24, wide 0.17.

Deliverable: `part3_practice.py` matching the hand calculations, then a real out-of-sample backtest.

Three Questions on the Cases

If you can answer these, the course has landed

Q1 How do you turn a directional forecast into a P&L?

**Strategy return =
 $\text{sign}(\text{signal}) \times \text{market}$**

Q2 What is the ultimate grade for the strategy?

**Out-of-sample P&L vs
buy & hold**

Q3 What happens to the bet when the band widens?

**It shrinks - you risk
less when unsure**

All three come from the strategy P&L, the buy-and-hold benchmark, and the conviction-sizing rule.

Three Pitfalls to Avoid

Where the final step slips

Common mistakes

- "A good scoreboard finishes the project."
- "Bet the same size no matter the conviction."
- "Six good days prove the strategy."

The reality

- Grade the strategy's out-of-sample P&L too.
- Size by edge, conviction, and band width.
- Test longer, with costs, before you believe it.

Two Calculations to Remember

Part 3 applications in one screen

1

Grade the strategy's out-of-sample P&L

Strategy = sign(signal) x market gave +2.70% versus +1.10% for buy-and-hold, with a higher per-day Sharpe and a 4-of-6 hit rate. The P&L, not the RMSE, is the final grade of a trading study.

2

Size the bet by conviction and the band

Size = edge x P(favorable) x (1 - band). A wide band cut the bet from 0.24 to 0.17 - about 30% smaller - purely because the model was less sure. Honest doubt becomes safer risk.

3

That is the whole course, in one trade

A point sets the hedge, a regime gates it, a band sizes it - all tested out of sample first. Match the model to the decision, test everything, and let uncertainty size the bet.

The core course is complete - thank you, and good luck with the final project.